Wild-place metaplectic character transfer in degree two: Toral positivity and the nonabelian gluing problem

Draft 6 — standalone companion to Part IV

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Abstract

We study irreducible-character integrality for the scalar-central odd-degree metaplectic cover of a quaternion division algebra at a wild place. The general result is a uniform quadratic toral permutation-character theorem: for every quadratic E/F in odd residue characteristic and every odd n with $\mu_n \subset F$, the discriminant ratio $\Delta(t)/\Delta(t^n)$ is a finite permutation character of $E^\times/F^\times$. The main new result concerns the running example $F = \mathbb{Q}_3(\zeta_3)$, $n = 3$, D quaternion. We prove that the division-side transform L_D of Part III carries the trivial character of $D^\times$ to the character of a single genuine irreducible representation τ of the cover, of dimension 27 (Target A of Draft 0), and we describe τ: its restriction to the pro-3 part of the cover is induced from a canonical genuine character of the covered third congruence subgroup. The proof has three layers: a reduction, via Brauer's characterization and the toral theorem, to the pro-3 Sylow subgroup; a structural analysis of the good elements of the congruence filtration which turns the question into a finite combinatorial problem about 72 vectors in $\mathbb{F}_3^5$ that has exactly two mirror-image solutions; and an explicit cocycle model of the cover, in which the one remaining identity (multiplicativity of the canonical star section on the sixth congruence subgroup) is proved by a short computation with cubic Hilbert symbols. The same holds for every irreducible representation of $D^\times$ trivial on the third congruence subgroup and compatible with the cover (central character trivial on μ_3): the transform of its character is the character of a genuine irreducible representation whose restriction to the pro-3 part is a sum of irreducible representations induced from the covered third congruence subgroup, one for each orbit of genuine constituents of a "cube-root transform" of $\rho|_{K^1}$ (part of Target B). For an arbitrary row we prove the reduction to the pro-3 Sylow subgroup and a sufficient criterion in terms of a cube-root transform of Θ_ρ. Finally, the p-th power identity and the multiplicativity of the canonical star section on the $2d$-th congruence subgroup are proved for every odd prime p and every $F \supset \mu_p$ (part of Target C); what is not uniform is identified.

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1 Purpose, scope, and status

This paper separates the wild division-side character problem from the conditional global trace comparison in Part IV of the larger series. Draft 0 proved the general toral theorem (Section 4),

worked out the three quadratic tori of the running example (Section 6), proved the positivity–integrality criterion (Section 3), and formulated three targets:

Target A
For $F = \mathbb{Q}_3(\zeta_3)$, D quaternion and $n = 3$, decide whether $L_D\mathbf{1}$ is the character of a single 27-dimensional genuine irreducible representation of $\widetilde{H}$ with the prescribed central character.

Target B
For every ordinary irreducible ρ with a fixed compatible central character, prove that all coefficients $a_{\rho\tau}$ are integral, or exhibit a nonintegral one.

Target C
Extend Target B to all odd n, all F of odd residual characteristic with $\mu_n \subset F$, and quaternion D/F.

Drafts 1–6 settle Target A affirmatively (Theorem 11.1), settle Target B for all representations trivial on K^3 and their norm twists (Theorems 12.5, 12.7), prove the reduction theorem for every row (Theorem 12.1) and a sufficient criterion for a row (Proposition 12.4), and prove the two cocycle-level ingredients of Target C uniformly in p (Theorems 13.2, 13.3). The dependencies of the main theorem are listed in Remark 11.2.

Labels. [**P**] proved here; [**E**] established elsewhere and used as a black box; [**O**] outlined; [**C**] conjectural. The new label [$\mathbf{P}^{\mathrm{c}}$] marks a statement whose proof uses, besides hand arguments, an exact finite computation (linear algebra over $\mathbb{F}_3$, or exact arithmetic in a number field) that is *exhaustive* over an explicitly described finite set. Every such computation is described in the text, and the scripts are listed in Appendix B. No statement labeled [$\mathbf{P}^{\mathrm{c}}$] depends on floating-point arithmetic or on sampling. Sampled computations occur in the paper only as corroboration of statements that are proved by hand, and are identified as such where they are mentioned.

Statement	Source	Status
Unitary local L_D and identity factor	Part III, Draft 9	[**E**]
Heisenberg commutator and degree on maximal tori	Part I, Draft 8	[**E**]
Quadratic toral permutation-character theorem	Theorem 4.1	[**P**]
Three exact cubic wild field-torus calculations	Section 6	[**P**]
All-row positivity–integrality equivalence	Proposition 3.1	[**P**]
Structure of good elements in the congruence filtration	Section 7	[**P**]
Reduction of Target A to the pro-3 Sylow subgroup	Theorem 8.5	[**P**]
Finite model; the defect function; uniqueness of the design	Section 9	[$\mathbf{P}^{\mathrm{c}}$]
Explicit cocycle model (Kubota–Kazhdan–Patterson)	Lemma 10.1	[**E**]/[**P**]
Cube identity; star section on K^6; proof of (H1)	Section 10	[**P**]
$L_D\mathbf{1}$ is a genuine irreducible character (Target A)	Theorem 11.1	[$\mathbf{P}^{\mathrm{c}}$]
Reduction theorem for every row	Theorem 12.1	[**P**]
Cube-root transform; sufficient criterion for a row	Proposition 12.4	[**P**]
All rows trivial on K^2, and norm twists (part of Target B)	Theorem 12.5	[$\mathbf{P}^{\mathrm{c}}$]
All rows trivial on K^3 (part of Target B)	Theorem 12.7	[$\mathbf{P}^{\mathrm{c}}$]
Target B for level ≥ 3	Section 12.5	[**O**]
p-th power identity and (H1) for every odd p (part of Target C)	Section 13	[**P**]
Target C: Brauer reduction; finite design	Remark 13.4	[**O**]/[**C**]

2 Division-side character transform and its normalization

Let F be nonarchimedean, D/F a division algebra of prime degree r, $H = D^\times$, and $\widetilde{H}$ its specified cover. Set $Z = F^\times$ and $\widetilde{Z} = p^{-1}(Z)$, equipped with compatible Haar measures on $H/Z \simeq \widetilde{H}/\widetilde{Z}$. The Part III transform is

$$(L_D\Phi)(z\,t^*) = b_D\,\widetilde{\theta}(z)\frac{\Delta(t)}{\Delta(t^n)}\Phi(t), \qquad b_D = |n|_F^{-(r-1)/2}, \tag{1}$$

for regular t, $z \in \widetilde{Z}$; it vanishes at nongood regular lifts. Here $t^* = \tilde{t}^{\,n}$ is the star of any lift $\tilde{t}$ of t, and

$$\Delta(t) = \left|\det(1 - \mathrm{Ad}(t) \mid \mathfrak{h}/\mathfrak{t})\right|_F^{1/2} = \prod_{i<j}\left|\frac{(\lambda_i - \lambda_j)^2}{\lambda_i\lambda_j}\right|_F^{1/2}. \tag{2}$$

This equals Part I's $|D(t)|^{1/2}|\,\mathrm{Nrd}(t)|_F^{-(r-1)/2}$. Since the power map on a field torus modulo scalars has index $|n|_F^{-(r-1)}$, Weyl integration gives b_D^2 equal to this index. The factor b_D is independent of *common* rescalings of the two identified quotient Haar measures. The ratio of discriminants extends at central elements to $|n|_F^{-r(r-1)/2}$ (Part III, Lemma 3.3), hence

$$(L_D\Theta_\rho)(1) = c_D\,\dim\rho, \qquad c_D = b_D|n|_F^{-r(r-1)/2} = |n|_F^{-(r^2-1)/2}. \tag{3}$$

The identity factor is pointwise and does not depend on Haar measures. Part III, Theorem 9.1, separately gives the adjoint orbital-integral/test-function transfer. Test functions transform by $\widetilde{\theta}^{-1}$, not by $\widetilde{\theta}$.

2.1 Integral-matrix criterion

Choose $\widetilde{\theta}$ first and put $\theta(a) = \widetilde{\theta}(a^*)$. Part III, Theorem 4.3 and Corollary 5.1, show that L_D is a unitary *isomorphism* of invariant L^2 spaces and that

$$L_D\Theta_\rho = \sum_\tau a_{\rho\tau}\Theta_\tau, \qquad a_{\rho\tau} = \langle L_D\Theta_\rho, \Theta_\tau\rangle, \tag{4}$$

with each row and column finite. The characters on either side are orthonormal bases in their fixed-central-character spaces.

Proposition 2.1 ([**E**] Integral-lattice criterion, Part III)**.** *The transform L_D permutes the irreducible characters with positive signs if and only if every $a_{\rho\tau}$ in (4) is an integer.*

Proof. An integral unit vector in ℓ^2 is a signed basis vector; unitarity gives injectivity and surjectivity because the inverse is the adjoint of a unitary map. The value at identity in (3) is positive and fixes the sign. This is Part III, Corollary 5.2, with its converse. □

Remark 2.2 (The actual obstruction)**.** For wild v, Part III does not prove $a_{\rho\tau} \in \mathbb{Z}$. The tame Adams argument works with a finite quotient Q and a central subgroup C, under the hypothesis $\gcd(n, |Q/C|) = 1$ (Part III, Lemma 7.5). At a wild place that hypothesis need not hold. For the simplest illustration take $Q = C_p$, $C = 1$, $n = p$: then the Adams operation sends every linear character to the trivial character. This does not prove a nonintegral coefficient for L_D; it only shows why that particular proof method fails. The present paper replaces the Adams argument, in the running example, by Brauer's characterization of characters and an explicit analysis of the pro-3 part of the cover.

3 A useful global criterion: positivity of every row

Proposition 3.1 ([**P**] Positivity versus integrality)**.** *With the unitary character matrix* $(a_{\rho\tau})$ *of* (4)*, the following are equivalent:*

(a) L_D *permutes irreducible characters with positive signs;*

(b) every $a_{\rho\tau}$ *is an integer;*

(c) every $a_{\rho\tau}$ *is a nonnegative real number;*

(d) every $L_D\Theta_\rho$ *is positive definite on* $\widetilde{H}$*.*

The implication from positive definiteness to integrality requires every *row; positivity of a single row does not force integrality or irreducibility for that row.*

Proof. By Part III, each row and column of A has ℓ^2 norm one. An integer row is a signed basis vector, and its sign is positive by the identity value (3). If all entries are nonnegative, orthogonal rows have disjoint supports. Every column meets exactly one row, and since its norm is one its nonzero entry is one; row norms then imply the permutation property. The equivalence of nonnegative character coefficients and positive definiteness follows from Peter–Weyl/GNS for a compact-mod-center group with fixed unitary central character. One row could instead be $(2^{-1/2}, 2^{-1/2}, 0, \ldots)$. □

Remark 3.2. For a *single* row the usable criterion is different and is the one we shall apply: a class function of ℓ^2 norm one with positive value at the identity is an irreducible character as soon as it is a *virtual* character (an integral combination of irreducible characters). Brauer's theorem reduces the virtual-character property to restrictions to elementary subgroups; Section 8 carries this out.

4 A general quadratic theorem

Let F be a finite extension of $\mathbb{Q}_p$ with p odd, residue field of cardinality q, and let $n \geq 1$ be odd with $\mu_n \subset F$. Let E/F be any quadratic field extension, with nontrivial Galois involution σ. Put

$$A = E^\times/F^\times, \qquad G = E^1 = \ker(N_{E/F} : E^\times \to F^\times).$$

Hilbert's Theorem 90 identifies $A \xrightarrow{\sim} G$ via $t \mapsto u = t/\sigma(t)$. The normalized absolute value on E is $|x|_E = q_E^{-v_E(x)}$, so $|x|_E = |N_{E/F}x|_F$.

For noncentral $t \in E^\times$ let

$$R_E(t) = \frac{\Delta(t)}{\Delta(t^n)}, \qquad \Delta(t) = \left|\det(1 - \mathrm{Ad}(t) \mid D/E)\right|_F^{1/2}.$$

The ratio extends continuously at scalar elements, with value $|n|_F^{-1}$.

Theorem 4.1 ([**P**] Quadratic toral permutation positivity)**.** *For any* $E/F, n$ *as above,* R_E *is the character of a finite-dimensional permutation representation of a finite quotient of* A*. In particular,*

$$R_E(t) = \sum_{\chi \in \widehat{A}} m_\chi \chi(t), \qquad m_\chi \in \mathbb{Z}_{\geq 0},$$

where only finitely many m_χ *are nonzero. This includes every wild case* $p \mid n$*. At the identity* $R_E(1) = |n|_F^{-1}$*.*

4.1 A radial valuation formula

Write $n = p^a m$ with $(m, p) = 1$, and put

$$t_\zeta = v_E(1-\zeta), \qquad G_k = \{u \in G : v_E(1-u) \geq k\} \quad (k \geq 1), \qquad G_0 = G.$$

If $a = 0$ then $R_E \equiv 1$, so assume $a \geq 1$.

Lemma 4.2. *For $u \in G$ and any $\zeta \in \mu_n$ whose prime-to-p component is nontrivial, $|1-\zeta u|_E = 1$. For $\zeta \in \mu_{p^a} \setminus \{1\}$,*

$$v_E(1-\zeta u) = \min\{v_E(1-u), t_\zeta\}.$$

Consequently, with $k = v_E(1-u) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$,

$$R_E(u) = q_E^{\frac{1}{2}\sum_{\zeta \in \mu_{p^a}\setminus\{1\}} \min(k, t_\zeta)}. \tag{5}$$

Proof. The reduction of $u \in E^1$ lies in a norm-one subgroup. In the unramified case its intersection with $k_F^\times$ is $\{\pm 1\}$, and in the ramified case the reduction of u is itself in $\{\pm 1\}$. A root of unity with nontrivial prime-to-p component has reduction in $k_F^\times \setminus \{\pm 1\}$, since n is odd, proving the first assertion.

For a nontrivial p-power root ζ, the ultrametric inequality gives the asserted minimum when the two valuations differ. If both equal t_ζ and $v_E(1-\zeta u) > t_\zeta$, then $u \equiv \zeta^{-1} \pmod{\mathfrak{p}_E^{t_\zeta+1}}$. Applying σ and using $\sigma(u) = u^{-1}$ gives $u^{-1} \equiv \zeta^{-1}$ to the same precision, hence $\zeta \equiv \zeta^{-1}$, a contradiction to $v_E(\zeta - \zeta^{-1}) = t_\zeta$ (as p is odd). Thus equality holds.

For $u = t/\sigma(t)$ we have $\Delta(t) = |N_{E/F}(1-u)|_F^{1/2} = |1-u|_E^{1/2}$. Factor $1-u^n = \prod_{\zeta \in \mu_n}(1-\zeta u)$ and use the two valuation statements. The identity follows for noncentral t and extends continuously to $u = 1$. At $u = 1$, the cyclotomic product $\prod_{\zeta \in \mu_{p^a}\setminus\{1\}}(1-\zeta) = p^a$ gives $R_E(1) = |p^a|_E^{-1/2} = |n|_F^{-1}$. □

Define for $k \geq 0$

$$M_k = q_E^{\frac{1}{2}\sum_{\zeta \in \mu_{p^a}\setminus\{1\}} \min(k, t_\zeta)}, \qquad M_0 = 1,$$

and put $T = \max_{\zeta \neq 1} t_\zeta$. All values M_k are positive integers because roots ζ, ζ^{-1} occur in pairs of equal valuation. We have $M_k = M_T$ for $k \geq T$.

4.2 The unramified filtration

Suppose E/F is unramified. Then $q_E = q^2$ and reduction of the norm-one group has order $q+1$; at every deeper level the trace-zero subgroup of the residue-field additive group has order q. Thus the filtration satisfies

$$[G : G_1] = q+1, \qquad [G : G_k] = (q+1)q^{k-1} \quad (k \geq 1).$$

Write $s_k = \#\{\zeta \in \mu_{p^a} \setminus \{1\} : t_\zeta \geq k\}$. Each s_k is even, and $M_k/M_{k-1} = q^{s_k}$. Set

$$c_k = \frac{M_k - M_{k-1}}{(q+1)q^{k-1}} \quad (1 \leq k \leq T).$$

For $k = 1$, $M_0 = 1$ and $q+1$ divides $q^{s_1} - 1$ because s_1 is even. For $k > 1$ with $s_k > 0$, M_{k-1} is divisible by q^{k-1}, since at least two roots contribute at least $k-1$ to its exponent. Again $q+1$ divides $q^{s_k} - 1$. Hence $c_k \in \mathbb{Z}_{\geq 0}$ (and $c_k = 0$ if $s_k = 0$).

By telescoping (5),

$$\boxed{R_E = \mathbf{1}_G + \sum_{k=1}^{T} c_k \, [G : G_k] \, \mathbf{1}_{G_k}.} \tag{6}$$

Here $[G : G_k]\,\mathbf{1}_{G_k}$ is precisely the permutation character of G on G/G_k.

4.3 The ramified filtration

Suppose E/F is ramified. Then $q_E = q$ and $t_\zeta = 2v_F(1-\zeta)$ is even. The norm-one group has filtration

$$[G : G_1] = 2, \qquad G_{2j} = G_{2j+1}, \qquad [G : G_{2j}] = 2q^j \quad (j \geq 1).$$

For completeness, write $E = F(\sqrt{d})$ with $v_F(d) = 1$. Hilbert 90 gives $u = (a + b\sqrt{d})/(a - b\sqrt{d})$; normalize a, b to be integral and not both divisible by the uniformizer of F. If a is a unit, scale it to 1 and write $c = b/a \in \mathcal{O}_F$: then the depth of $u - 1$ is $1 + 2v_F(c)$. Moreover, two such parameters c, c' define elements congruent modulo $\mathfrak{p}_E^{2j}$ precisely when $c \equiv c' \pmod{\mathfrak{p}_F^j}$, so $[G_1 : G_{2j}] = q^j$. If a is not a unit and b is, then $u \equiv -1$ modulo $\mathfrak{p}_E$. These observations give the displayed indices and the fact that positive depths are odd.

Define

$$c_1 = \frac{M_1 - 1}{2}, \qquad c_{2j} = \frac{M_{2j+1} - M_{2j-1}}{2q^j} \quad (1 \leq j \leq T/2).$$

The first coefficient is an integer because $M_1 = q^{(p^a-1)/2}$ is odd. For $j \geq 1$ write

$$\frac{M_{2j+1}}{M_{2j-1}} = q^{e_j}, \qquad e_j = \frac{1}{2}\sum_{\zeta \neq 1} (\min(2j+1, t_\zeta) - \min(2j-1, t_\zeta)) \in \mathbb{Z}_{\geq 0}.$$

If $e_j = 0$, then $c_{2j} = 0$. Otherwise some pair $\{\zeta, \zeta^{-1}\}$ has $t_\zeta \geq 2j$, so M_{2j-1} is divisible by q^{2j-1}, hence by q^j. Moreover $q^{e_j} - 1$ is even. Therefore $c_{2j} \in \mathbb{Z}_{\geq 0}$.

As all positive depths are odd, telescoping now gives

$$\boxed{R_E = \mathbf{1}_G + c_1\,[G : G_1]\mathbf{1}_{G_1} + \sum_{j=1}^{T/2} c_{2j}\,[G : G_{2j}]\,\mathbf{1}_{G_{2j}}.} \tag{7}$$

Again each summand is the character of a permutation representation. Equations (6) and (7) prove Theorem 4.1.

5 From Fourier coefficients to covered-torus restrictions

Lemma 5.1 ([**P**] Fourier test for restrictions to covered field tori)**.** *Let E/F be any maximal field in D, with r prime and $(r, n) = 1$, and put $A_E = E^\times/F^\times$. The function*

$$R_E(t) = \frac{\Delta(t)}{\Delta(t^n)}$$

extends continuously at scalars, is a locally constant function on the compact abelian group A_E, and has a finite Fourier expansion

$$R_E(t) = \sum_{\chi \in \widehat{A_E}} m_\chi \chi(t), \qquad m_\chi = \int_{A_E} R_E(t)\overline{\chi(t)}\,dt. \tag{8}$$

Here the Haar measure of A_E has mass 1. Choose a compatible genuine central character $\widetilde{\theta}$ of $\widetilde{Z}$ and set $\theta(a) = \widetilde{\theta}(a^*)$. For any smooth character $\alpha : E^\times \to \mathbb{C}^\times$ with $\alpha|_{F^\times} = \theta$, and for each $\chi \in \widehat{A_E}$, let $\eta_{\alpha\chi}$ be the unique genuine irreducible representation of $\widetilde{E^\times}$ whose central character satisfies

$$\zeta_{\alpha\chi}(z\,t^*) = \widetilde{\theta}(z)\alpha(t)\chi(t) \qquad (z \in \widetilde{Z},\ t \in E^\times).$$

Define a torus-level class function $\mathcal{L}_E\alpha$ to equal $b_D\widetilde{\theta}(z)R_E(t)\alpha(t)$ on $zt^* \in C_E$ and to vanish outside C_E. Then, as a genuine class function on the covered torus*,*

$$\mathcal{L}_E\alpha = \sum_{\chi} m_\chi \, \Theta_{\eta_{\alpha\chi}}. \tag{9}$$

For an ordinary character Θ_ρ of $D^\times$, its actual L_D transform restricts to the sum of these $\mathcal{L}_E\alpha$ with the multiplicities in $\Theta_\rho|_{E^\times}$. Consequently $m_\chi \in \mathbb{Z}_{\geq 0}$ for every χ is sufficient for the restriction of every transferred ordinary irreducible character to this covered torus to be a genuine representation character. It is not sufficient for the full $\widetilde{D^\times}$ character to be a representation character.

Proof. The extension of R_E at scalars follows from Part III, Lemma 3.3. Since E/F is a degree-r field extension with $(r, n) = 1$, the nth power of a noncentral t stays noncentral. The compactness of A_E and local constancy imply that R_E factors through a finite quotient, giving (8).

By Part I the center of the covered torus is $C_E = \widetilde{Z} \cdot \{t^* : t \in E^\times\}$. The displayed formula for $\zeta_{\alpha\chi}$ is well-defined: the power map on A_E is injective, and on scalar overlaps the compatibility $\alpha(a) = \widetilde{\theta}(a^*)$ is precisely what is required. Such genuine central characters extend over C_E, and Stone–von Neumann gives a unique $\eta_{\alpha\chi}$. Its character vanishes off C_E, while on C_E it equals $b_D\zeta_{\alpha\chi}$ because its degree is b_D. The defining formula (1) for L_D has the identical vanishing and equals $b_D\widetilde{\theta}(z)R_E(t)\alpha(t)$ on zt^*. Comparing with (8) proves (9). Finally, restricting a finite-dimensional ordinary representation to $E^\times$ decomposes it into one-dimensional characters, all restricting to θ on $F^\times$; summing the displayed formula proves the last claim. □

6 The explicit cubic wild example

The following calculation verifies a *necessary representation-ring integrality test on every maximal torus*, including the two ramified quadratic field types. Sections 7–11 then settle the full-group statement.

Example 6.1 (The field, algebra, and cover)**.** Take

$$F = \mathbb{Q}_3(\zeta_3), \qquad \varpi = \zeta_3 - 1, \qquad 3 = -\zeta_3^{-1}\varpi^2, \qquad q_F = 3.$$

Thus $v_F(3) = 2$ and $|3|_F = 3^{-2} = 1/9$. The compatible covering parameters in [JC] and Part I are respectively $c = 1$ and $c_v = 2 \pmod 3$. Let D/F be the unique quaternion division algebra. With $E_0 = F(i)$, $i^2 = -1$, its unramified quadratic maximal subfield, we may write

$$D = E_0 \oplus E_0 j, \qquad j^2 = \varpi, \qquad jx = \sigma(x)j \quad (x \in E_0),$$

where σ is the nontrivial automorphism of E_0/F. Part I, Proposition 3.1, gives, for every quadratic maximal subfield E/F, the covered-torus commutator

$$[x, y]_{\widetilde{E^\times}} = (N_{E/F}x, N_{E/F}y)^2_{F,3}(x, y)^{-1}_{E,3}. \tag{10}$$

The center C_E of $\widetilde{E^\times}$ is the preimage of $F^\times E^{\times 3}$ (Part I, Proposition 3.5), and

$$[E^\times : F^\times E^{\times 3}] = 9, \qquad b_D = 3, \qquad c_D = 27.$$

By the Heisenberg-type Stone–von Neumann theorem ([JC], Proposition 2.4; Part I, Proposition 3.5), every genuine irreducible representation of $\widetilde{E^\times}$ has degree 3. This is *not* a degree calculation for representations of $\widetilde{D^\times}$.

6.1 The unramified quadratic torus

Put $E_0 = F(i)$ as above. Since $X^2 + 1$ is irreducible over $\mathbb{F}_3$, E_0/F is unramified and its residue field is $\mathbb{F}_9$. For $t = a + bi$ with $a, b \in F$ the reduced norm and trace give the elementary identity

$$R_{E_0}(a + bi) = \frac{|a^2 + b^2|_F}{|3a^2 - b^2|_F}. \tag{11}$$

Multiplying t by an element of $F^\times$, suppose $a, b \in \mathcal{O}_F$ and at least one is a unit. Then $a^2 + b^2$ is always a unit. If b is a unit, so is $3a^2 - b^2$. If $b = \varpi u$, then

$$3a^2 - b^2 = 3(a^2 + \zeta_3 u^2),$$

and a is a unit; the reduction of $a^2 + \zeta_3 u^2$ modulo ϖ is either 1 or 2, never 0. Hence

$$R_{E_0}(a + bi) = \begin{cases} 9, & b \in \varpi\mathcal{O}_F, \\ 1, & b \in \mathcal{O}_F^\times. \end{cases} \tag{12}$$

The finite residue quotient is

$$B_0 = k_{E_0}^\times / k_F^\times \simeq C_4.$$

The condition $b \in \varpi\mathcal{O}_F$ says exactly that the image of t in B_0 is the identity. Denote its four characters by $\chi_0 = 1, \chi_1, \chi_2, \chi_3$. The regular character of C_4 equals 4 at the identity and 0 elsewhere. Thus

$$\boxed{R_{E_0} = 1 + 2\,\mathrm{reg}_{C_4} = 3\chi_0 + 2\chi_1 + 2\chi_2 + 2\chi_3.} \tag{13}$$

This is a positive integral Fourier decomposition of degree $3 + 2 + 2 + 2 = 9$; it is stronger than merely checking that R_{E_0} takes integer values.

6.2 Both ramified quadratic tori

Every ramified quadratic extension of F can be written $E_d = F(j_d)$, $j_d^2 = d = u\varpi$ with $u \in \mathcal{O}_F^\times$. There are two isomorphism classes, distinguished by the square class of u. For $t = a + bj_d$,

$$R_{E_d}(a + bj_d) = \frac{|a^2 - db^2|_F}{|3a^2 + db^2|_F}. \tag{14}$$

Again normalize a, b to be integral and not both divisible by ϖ. The terms $3a^2$ and db^2 have different valuations whenever both a and b are units. A direct valuation calculation therefore gives

$$R_{E_d}(a + bj_d) = \begin{cases} 9, & a \in \mathcal{O}_F^\times, \quad b \in \varpi\mathcal{O}_F, \\ 3, & a, b \in \mathcal{O}_F^\times, \\ 1, & a \in \varpi\mathcal{O}_F, \quad b \in \mathcal{O}_F^\times. \end{cases} \tag{15}$$

For example, in the original cyclic algebra $d = \varpi$,

t	j	$1 + j$	$1 + \varpi j$
$R(t)$	1	3	9
$(L_D 1)(t^*)$	3	9	27

for the central-character normalization of Example 6.1. The three values illustrate why the unramified torus alone is not an adequate test.

There is a natural finite quotient

$$B_d = E_d^\times/(F^\times U_1) \simeq C_2 \times C_3,$$

where $U_1 \subset \mathcal{O}_{E_d}^\times$ consists of $a + bj_d$ with $a \in \mathcal{O}_F^\times$ and $b \in \varpi\mathcal{O}_F$; equivalently $U_1 = \mathcal{O}_F^\times(1 + \mathfrak{p}_{E_d}^2)$. The C_2 coordinate records the parity of the E_d-valuation, and on even elements the C_3 coordinate is $\bar{c} \in \mathbb{F}_3$ for $1 + cj_d$ with $c \in \mathcal{O}_F$ read modulo ϖ. The multiplication law

$$(1 + cj_d)(1 + c'j_d) = (1 + dcc')\left(1 + \frac{c + c'}{1 + dcc'}j_d\right)$$

shows that $\bar{c}$ is additive, and $j_d^2 = d \in F^\times$ provides the order-two factor.

Write ϵ for the nontrivial character of C_2 and ξ for a character of order 3 of C_3. The identity (15) says that R_{E_d} has values 9 at the identity of B_d, 3 at its two nonidentity elements in C_3, and 1 on its three odd elements. Consequently

$$\boxed{\begin{aligned} R_{E_d} &= 1 + \mathrm{reg}_{C_2} + \mathrm{reg}_{C_2 \times C_3} \\ &= 3 \cdot 1 + 2\epsilon + \xi + \xi^2 + \epsilon\xi + \epsilon\xi^2. \end{aligned}} \tag{16}$$

Here the regular representations are inflated from the indicated quotients. All coefficients are nonnegative integers and sum to $3 + 2 + 1 + 1 + 1 + 1 = 9$. This applies to *both* ramified quadratic isomorphism classes, independently of u.

Proposition 6.2 ([**P**] All quadratic torus restrictions pass an integral-character test)**.** *In the wild example $F = \mathbb{Q}_3(\zeta_3)$, $r = 2$, $n = 3$, let D/F be the quaternion division algebra. The function R_E is the character of a nine-dimensional permutation representation of a finite quotient of $E^\times/F^\times$, for every quadratic maximal subfield $E \subset D$. More precisely, for unramified E it is $1 \oplus 2\,\mathbb{C}[C_4]$, and for either ramified E it is $1 \oplus \mathbb{C}[C_2] \oplus \mathbb{C}[C_2 \times C_3]$. For a suitable genuine central character inducing the trivial ordinary central character, the restriction of $L_D\mathbf{1}$ to $\widetilde{E^\times}$ is therefore an* actual genuine representation character *of total dimension* 27*, namely*

E/F	*multiplicities of three-dimensional genuine torus irreducibles*
unramified	$(3, 2, 2, 2)$
ramified, either type	$(3, 2, 1, 1, 1, 1)$.

These multiplicities are invariant under the nontrivial F-automorphism of each E.

Proof. A p-adic field of odd residue characteristic has four square classes; hence F has exactly three quadratic field extensions, one unramified and two ramified. Every quadratic field extension embeds in the quaternion division algebra. Equations (13) and (16) prove the first assertion. Lemma 5.1, together with the degree-three torus Stone–von Neumann theorem, converts those Fourier coefficients into the displayed multiplicities; their weighted dimensions are $3 \cdot 9 = 27$. For the trivial ordinary character one can choose $\widetilde{\theta}$ with $\widetilde{\theta}(a^*) = 1$: the odd-$n$ scalar star kills $\mu_3 \subset F^\times$ (Part I, Section 3), and a faithful character of μ_3 extends across the abelian scalar central group modulo the star image. The nontrivial Galois automorphism acts by inversion on the respective finite quotients' character parameters (with the C_2 factor fixed), so the displayed multiplicities are invariant. □

Remark 6.3 (Standing conventions for the rest of the paper)**.** From here on $F = \mathbb{Q}_3(\zeta_3)$, $n = 3$, D is the quaternion division algebra, $E_0 = F(i)$, and j, ϖ are as in Example 6.1. We write $\mathcal{O}_D$ for

the maximal order, $\mathfrak{P} = \mathfrak{P}_D = j\mathcal{O}_D$ for its maximal ideal, v_D for the valuation with $v_D(j) = 1$ (so $v_D(\varpi) = 2$ and $v_D(3) = 4$), and

$$K^m = 1 + \mathfrak{P}^m, \qquad U_m = F^\times K^m, \qquad \widetilde{U}_m = p^{-1}(U_m) \subset \widetilde{H} \qquad (m \geq 1).$$

Part III's log records that the standard scalar cocycle is trivial; accordingly $\widetilde{Z} \simeq \mu_3 \times F^\times$ and a genuine central character has the form $\widetilde{\theta}(\zeta, a) = \zeta\theta_0(a)$. For $\rho = \mathbf{1}$ the compatibility $\widetilde{\theta}(a^*) = 1$ forces $\theta_0^3 = 1$; we fix one such θ_0 (there are 81; by Remark 8.6 the choice is immaterial) and note that $\widetilde{\theta}(\widetilde{Z}) = \mu_3$. The central character θ_ρ of any ρ to which L_D applies is trivial on μ_3, since $\theta_\rho(\zeta) = \widetilde{\theta}(\zeta^*) = \widetilde{\theta}(1) = 1$.

7 The pro-3 structure of K^1 and the cube map

The whole wild phenomenon lives on the pro-3 group

$$Q := K^1F^\times/F^\times \;\simeq\; K^1/U_F^{(1)}, \qquad Q_m := \text{image of } K^m \quad (m \geq 1),$$

where $U_F^{(k)} = 1 + \mathfrak{p}_F^k$ and $K^1 \cap F^\times = U_F^{(1)}$. Since $U_F^{(k)} = F \cap K^{2k}$, the successive quotients are

$$Q_m/Q_{m+1} \simeq \begin{cases} \mathbb{F}_9, & m \text{ odd}, \\ \mathbb{F}_9/\mathbb{F}_3, & m \text{ even}, \end{cases} \qquad [Q : Q_6] = 3^8, \quad [Q : Q_3] = 27, \quad [Q_3 : Q_6] = 3^5. \tag{17}$$

Every element of $D^\times \setminus F^\times$ is regular and lies in a unique maximal subfield; in particular, Q is torsion-free, because $E^\times/F^\times$ has no 3-torsion for quadratic E (Part I, Lemma 3.4) and no 2-torsion inside a pro-3 group. The group $D^\times/F^\times K^1$ has order 8; the subgroup

$$P_0 := \langle 1 + i,\ j\rangle F^\times/F^\times \simeq D_8 \tag{18}$$

is a complement to Q in $D^\times/F^\times$ (indeed $(1+i)^4 = -4$, $j^2 = \varpi$ and $j(1+i)j^{-1} = 2(1+i)^{-1}$ in $D^\times$), and every finite 2-subgroup of $D^\times/F^\times$ is conjugate into P_0 (Schur–Zassenhaus).

7.1 The cube map

Lemma 7.1 ([P]). *(a) For $k \geq 3$ the cube map $K^k \to K^{k+4}$ is bijective.*

(b) The cube map $Q \to Q$ is injective.

(c) Let $t \in Q_1 \setminus Q_2$ have leading coefficient $\bar{w} \in \mathbb{F}_9^\times$ (that is, $t = 1 + w\pi + \cdots$ with $\pi = j$). Then $t^3 \in Q_3 \setminus Q_4$ with leading coefficient $\bar{w}^5$; in particular every nonzero leading coefficient occurs.

(d) $\mathrm{Cubes}(Q_2) = Q_6$.

(e) The set $\mathrm{Good}(Q)$ *of images of good elements is the set of cubes, and*

$$\mathrm{Good}(Q) = G_3 \sqcup Q_6, \qquad G_3 := \mathrm{Cubes}(Q_1 \setminus Q_2) \subset Q_3 \setminus Q_4.$$

Thus a good element of Q has depth exactly 3 or depth at least 6.

(f) G_3 is stable under multiplication by Q_6.

(g) $V := Q_3/Q_6$ is an elementary abelian group of order 3^5, and the image $\bar{G}_3$ of G_3 in V has exactly 72 elements, with $\bar{G}_3 = -\bar{G}_3$; it spans V.

Proof. (a) For $u \in \mathfrak{P}^k$, $(1+u)^3 = 1 + 3u + 3u^2 + u^3$ with $3u \in \mathfrak{P}^{k+4}$, $3u^2 \in \mathfrak{P}^{2k+4}$, $u^3 \in \mathfrak{P}^{3k}$; for $k \geq 3$ both error terms lie in $\mathfrak{P}^{k+5}$, and $u \mapsto 3u$ is a bijection $\mathfrak{P}^k/\mathfrak{P}^{k+1} \to \mathfrak{P}^{k+4}/\mathfrak{P}^{k+5}$. Successive approximation gives surjectivity; injectivity follows from (b) and $\mu_3 \cap K^3 = 1$.

(b) If $t^3 = t'^3$ in Q with $t \neq 1$, both t, t' lie in the torus of the regular element t^3 and $(t'/t)^3 \in F^\times$, hence $t'/t \in F^\times$ by Part I, Lemma 3.4.

(c) With $t = 1 + u$, $u = w\pi + \cdots$: $3u \in \mathfrak{P}^5$, $3u^2 \in \mathfrak{P}^6$ and $u^3 \equiv w\sigma(w)w\pi^3 = \mathrm{N}(\bar{w})\bar{w}\,\pi^3 \pmod{\mathfrak{P}^4}$, since $\pi a = \sigma(a)\pi$ for $a \in E_0$; and $\mathrm{N}(\bar{w})\bar{w} = \bar{w}^5$, with 5 prime to 8.

(d) Cubes of K^2 lie in K^6. Conversely every element of Q_6 is regular or trivial and lies in some torus $A = E^\times \cap F^\times K^1/F^\times$. For $E = E_0$ (so $E_0 \cap K^6 = U^{(3)}_{E_0}$): $U^{(4)}_{E_0}$ consists of cubes of $U^{(2)}_{E_0}$ (cubing $U^{(k)}_{E_0} \to U^{(k+2)}_{E_0}$ is bijective for $k \geq 2$, as $e(E_0/\mathbb{Q}_3) = 2$), and $U^{(3)}_{E_0}/U^{(4)}_{E_0} \simeq \mathbb{F}_9$ is covered by $U^{(3)}_F$ (the $\mathbb{F}_3$-part) together with the cubes $(1 + \varpi w)^3 \equiv 1 + \varpi^3(w^3 - w)$, whose leading coefficients fill the Artin–Schreier image $\mathbb{F}_3 i$. For ramified E (so $E \cap K^6 = U^{(6)}_E$, $e(E/\mathbb{Q}_3) = 4$): $U^{(7)}_E$ consists of cubes of $U^{(3)}_E$, and $U^{(6)}_E/U^{(7)}_E \simeq \mathbb{F}_3$ is covered by $U^{(3)}_F$. In both cases the cube root lies in $E \cap K^2$.

(e) The good elements of H are $F^\times \cdot \{\text{cubes}\}$ (Part I, Proposition 3.5(a), applied in each torus), so $\mathrm{Good}(Q) = \mathrm{Cubes}(Q)$. By (c), cubes of depth-one elements have depth exactly 3; by (d), cubes of Q_2 exhaust Q_6.

(f) Fix $t \in Q_1 \setminus Q_2$ with leading coefficient a and put $\tau = t - 1$. For $u \in \mathfrak{P}^k$ one has $(t(1+u))^3 = t^3(1 + L(u) + O(\mathfrak{P}^{2k}))$ with $L = 1 + \mathrm{Ad}(t^{-1}) + \mathrm{Ad}(t^{-2})$, and $L(u) \equiv (\mathrm{Ad}(t^{-1}) - 1)^2 u \equiv [[u, \tau], \tau]$ $(\mathrm{mod}\ \mathfrak{P}^{k+3})$ (ring commutators), because $3u \in \mathfrak{P}^{k+4}$. For $u = b\pi^3$ the double commutator is $-aD\pi^5$ with $D = b\sigma(a) - a\sigma(b)$, which vanishes iff $b \in \mathbb{F}_3 a$. Hence cubing maps $W := t \cdot (1 + \mathbb{F}_3 a\pi^3 + \mathfrak{P}^4)$ into $t^3 Q_6$. Cubing is a local homeomorphism near regular elements (its differential L is invertible, its determinant on $\mathfrak{d}/\mathfrak{f}$ having absolute value $|3|_F\, R(t)^{-2} = 3^{-4}$ for depth-one t), so the image of W is open in $t^3 Q_6$, and its measure is $\mu(W) \cdot 3^{-4} = 3^{-4} \cdot 3^{-4} = 3^{-8} = \mu(Q_6)$. Therefore cubing maps W onto $t^3 Q_6$, i.e. $t^3 Q_6 \subset G_3$. (The finite verification in Appendix B, script 2, confirms this statement in Q/Q_7.)

(g) $[K^3, K^3] \subset K^6$ and $(K^3)^3 \subset K^7$, so V is elementary abelian of order $[Q_3 : Q_6] = 3^5$. By (f) and the Jacobian computed in (f), $\mu(G_3) = \frac{8}{9} \cdot 3^{-4}$, which is one third of $\mu(Q_3 \setminus Q_4)$; hence $|\bar{G}_3| = \frac{1}{3} \cdot 216 = 72$. The explicit description is in Proposition 7.2; spanning and the symmetry $\bar{G}_3 = -\bar{G}_3$ are read off from it. □

7.2 Coordinates on V and the set $\bar{G}_3$

Write $\mathfrak{P}^3 = \varpi\mathcal{O}_{E_0} j \oplus \varpi^2 \mathcal{O}_{E_0}$ and $\mathfrak{P}^6 = \varpi^3 \mathcal{O}_D$. Since $(1+X)(1+Y) \equiv 1 + X + Y$ modulo $\mathfrak{P}^6$ for $X, Y \in \mathfrak{P}^3$, the map $1 + X \mapsto X$ identifies K^3/K^6 with the additive group $\mathfrak{P}^3/\mathfrak{P}^6$, and writing

$$X \equiv \varpi(A + B\varpi)j + \varpi^2 C, \qquad A, B, C \in \mathbb{Z}_9 = W(\mathbb{F}_9),$$

gives coordinates $(\bar{A}, \bar{B}, \bar{C}) \in \mathbb{F}_9^3$ (levels $3, 5, 4$ respectively). The scalars $U^{(2)}_F$ contribute exactly the $\mathbb{F}_3$-line in the C-coordinate. Hence

$$V = Q_3/Q_6 \simeq \mathbb{F}_9 \oplus (\mathbb{F}_9/\mathbb{F}_3) \oplus \mathbb{F}_9 \ni (\bar{A},\ \bar{C} \bmod \mathbb{F}_3,\ \bar{B}), \tag{19}$$

and we use the $\mathbb{F}_3$-basis $(A_{\mathrm{re}}, A_{\mathrm{im}}, C_{\mathrm{im}}, B_{\mathrm{re}}, B_{\mathrm{im}})$ in the tables.

Proposition 7.2 ([**P**^{c}] The good depth-three classes)**.** *With* $\mathrm{N}(a) = a\bar{a}$ *and* $\mathrm{Tr}(a) = a + \bar{a}$ *on* $\mathbb{F}_9$*,*

$$\bar{G}_3 = \Big\{ (a,\ x_4,\ y + \mathrm{N}(a)(\mathrm{Tr}(a\bar{y}) + x_4^2 - 1)a) \ :\ a \in \mathbb{F}_9^\times,\ x_4 \in \mathbb{F}_3 i,\ y \in \mathbb{F}_9 \Big\}, \tag{20}$$

a set of 72 *elements forming* 36 *lines through* 0 *(each line is* $\bar{G}_3 \cap \bar{A}_E$ *for the ramified tori* E *whose image in* V *is a fixed plane). Explicitly, for* $a \in \mathbb{F}_9^\times$ *and* $z \in \mathbb{F}_3\, i\bar{a}^{-1}$,

$$\bar{G}_3 = \{(a, 0, z - \mathrm{N}(a)a)\} \cup \{(a, \pm i, z + \mathrm{N}(a)a)\}.$$

Proof. For a ramified torus $E = F(j')$ with $j' = c\pi + c_2\pi^2 + c_3\pi^3 + \cdots$ pure (so $\mathrm{Tr}(c_2) = 0$, $j'^2 = u'\varpi$ with $\bar{u}' = \mathrm{N}(c)$), the cubes of depth-one elements $1 + j'w$, $w \equiv w_0 \in \mathcal{O}_F^\times$, are $\equiv 1 + (d'w_0^3 - \varpi^2 w_0)j'$ modulo $\mathfrak{P}^6$ and scalars, where $w \equiv w_0 \in \mathcal{O}_F^\times$. Reading off coordinates gives the vectors $\pm(\mathrm{N}(c)c, \mathrm{N}(c)c_2, \mathrm{N}(c)c_3 + (\bar{u}_1' - 1)c)$ with $\bar{u}_1' = \mathrm{Tr}(c\bar{c}_3) + c_2^2$, which after the substitution $a = \mathrm{N}(c)c$, $y = \mathrm{N}(c)c_3$ is (20). The map $y \mapsto y + \mathrm{N}(a)\,\mathrm{Tr}(a\bar{y})a$ has kernel $\mathbb{F}_3 a$ and image $\mathbb{F}_3\, i\bar{a}^{-1}$, giving the second description and the count $8 \cdot 3 \cdot 3 = 72$. The formula was verified by exact enumeration of the 648 depth-one classes modulo $K^4 U_F^{(1)}$ (Appendix B, script 1). □

7.3 The transferred trivial character on $\widetilde{U}_1$

Lemma 7.3 (**[P]**). *Put* $\Phi := L_D\mathbf{1}|_{\widetilde{U}_1}$ *and, for a good* $g \in U_1$, *write* $g = zt^3$ *with* $z \in F^\times$ *and* $t \in K^1$ *(*$t = 1$ *if* g *is central); its* star lifts *are the three elements* $\zeta\tilde{z}\,t^*$, $\zeta \in \mu_3$. *Then*

$$\Phi(\zeta\tilde{z}\,t^*) = \begin{cases} 27\,\zeta\,\widetilde{\theta}(\tilde{z}), & t \in F^\times K^2 \ (\textit{i.e. } g \in U_6), \\ 9\,\zeta\,\widetilde{\theta}(\tilde{z}), & t \in F^\times K^1 \setminus F^\times K^2 \ (\textit{i.e. } g \in F^\times G_3), \end{cases} \tag{21}$$

and Φ *vanishes at all lifts of non-good elements. In particular* Φ *is supported on* $\widetilde{U}_3$.

Proof. By (1), $\Phi(\tilde{z}\,t^*) = 3\widetilde{\theta}(\tilde{z})R_E(t)$ with $E = F(t)$. If $t \in F^\times K^2$ and $E = E_0$, then $t \in F^\times U_{E_0}^{(1)}$ and $R = 9$ by (12); if E is ramified, $t \in F^\times U_E^{(2)}$, which is the case $a \in \mathcal{O}_F^\times$, $b \in \varpi\mathcal{O}_F$ of (15), so $R = 9$. If $t \in F^\times K^1 \setminus F^\times K^2$, then E is ramified ($E_0 \cap K^1 \subset K^2$) and $t \in F^\times(U_E^{(1)} \setminus U_E^{(2)})$, the case a, b units, so $R = 3$. Good elements of U_1 are $F^\times \cdot \mathrm{Cubes}(K^1)$, whose images in Q are $G_3 \sqcup Q_6$ by Lemma 7.1(e). □

8 Reduction to the pro-3 Sylow subgroup

8.1 A finite quotient

Let $\phi := L_D\mathbf{1}$. By Part III the row of ϕ in (4) is finite: $\phi = \sum_{\tau \in S} a_\tau \Theta_\tau$ with S finite. Each $\tau \in S$ is a finite-dimensional smooth representation with central character $\widetilde{\theta}$, so $\ker\tau$ is an open normal subgroup of $\widetilde{H}$ with $\ker\tau \cap \widetilde{Z} = \ker\widetilde{\theta}$. Put

$$N := \widetilde{U}_1 \cap \bigcap_{\tau \in S} \ker\tau \cap N_6, \tag{22}$$

where $N_6 \subset \widetilde{U}_6$ is the open normal subgroup introduced in Definition 9.1 below (for the present section one may take $N_6 = \widetilde{U}_m$ for any large m). Then N is open and normal in $\widetilde{H}$, $N \cap \widetilde{Z} = \ker\widetilde{\theta}$, and

$$\Gamma := \widetilde{H}/N \quad \text{is finite}, \qquad T := \widetilde{Z}N/N \simeq \widetilde{Z}/\ker\widetilde{\theta} = \mu_3 \text{ is central}, \qquad \Gamma_0 := \widetilde{U}_1/N.$$

Γ_0 is a normal 3-subgroup of index 8 with $\Gamma/\Gamma_0 \simeq D_8$, hence the unique Sylow 3-subgroup; ϕ and $\Phi = \phi|_{\widetilde{U}_1}$ are class functions on Γ and Γ_0. For a maximal subfield E write $T_E := \widetilde{E^\times}N/N \subset \Gamma$.

Lemma 8.1 ([**P**] Toral restrictions descend)**.** *For every maximal subfield E, $\phi|_{T_E}$ is the character of a representation of T_E.*

Proof. By Proposition 6.2, $\phi|_{\widetilde{E^\times}}$ is the character of a (finite-dimensional, genuine) representation W. Since $\phi(gm) = \phi(g)$ for $m \in N \cap \widetilde{E^\times}$, in particular $\Theta_W(m) = \Theta_W(1) = \dim W$, so m acts trivially on W. □

8.2 Two-groups

Lemma 8.2 ([**P**] The star section on 2-groups)**.** *Let $\hat{P}_0 \subset \widetilde{H}$ be the preimage of $P_0F^\times$ and $\Gamma_{P_0} := \hat{P}_0N/N$. Then $\Gamma_{P_0} \simeq \mu_3 \times D_8$, the elements of 2-power order form the unique complement $\tilde{P}_0 \simeq D_8$ to μ_3, and the map $u \mapsto \sigma(u) := (u^{1/3})^*$ ($u^{1/3}$ the cube root in the group P_0 of order 8) is the inverse of the projection $\tilde{P}_0 \to P_0$. Consequently*

$$\phi|_{\Gamma_{P_0}} = 3\,\psi_P + 3\sum_\pi (\dim \pi)\,\Theta_\pi, \tag{23}$$

where ψ_P is the genuine linear character trivial on $\tilde{P}_0$ and π runs over the genuine irreducible representations of Γ_{P_0}; in particular $\phi|_{\Gamma_{P_0}}$ is a character.

Proof. $\hat{P}_0 \cap \widetilde{U}_1 = \widetilde{Z}$ because $P_0 \cap Q = 1$, so $\hat{P}_0 \cap N = \ker\widetilde{\theta}$ and Γ_{P_0} is a central extension of D_8 by $T = \mu_3$. Since $|D_8|$ is prime to 3, $H^2(D_8, \mu_3) = H^1(D_8, \mu_3) = 0$: the extension splits and the complement is unique; it consists of the elements of 2-power order. On each cyclic subgroup of P_0 the star map is a homomorphism (it is a homomorphism on every covered torus, as $(\tilde{x}\tilde{y})^3 = \tilde{x}^3\tilde{y}^3[\tilde{y},\tilde{x}]^3$ with $[\tilde{y},\tilde{x}] \in \mu_3$), so $\sigma(u)^8 = 1$ and $\sigma(u) \in \tilde{P}_0$ is the unique element of the complement over u. Every nontrivial $u \in P_0$ is of order 2 or 4 in a quadratic torus and has $R_E(u) = 1$ by (12)–(15) (the cases b a unit, resp. $a \in \varpi\mathcal{O}_F$), so by (1) $\phi(\zeta\tilde{z}\sigma(u)) = 3\zeta\widetilde{\theta}(\tilde{z})$ for $u \neq 1$ and $= 27\zeta\widetilde{\theta}(\tilde{z})$ for $u = 1$. That is $3\psi_P + 24\delta$, with δ the genuine delta function at μ_3, and $8\delta = \sum_\pi(\dim\pi)\Theta_\pi$. □

Lemma 8.3 ([**P**] Centralizers of 2-elements)**.** *Let $y \in \Gamma$ be an element of 2-power order whose image in $\Gamma/\Gamma_0 \simeq D_8$ is nontrivial, $\eta \in D^\times$ a representative of that image in $D^\times/F^\times$ and $E_y = F(\eta)$. (For $\rho = \mathbf{1}$ every nontrivial 2-element qualifies, T being μ_3; in Theorem 12.1 the central $\mu_2 \subset T$ is excluded, as its centralizer is all of Γ_0.)*

(a) $C_{\Gamma_0}(y) \subset T_{E_y}$.

(b) If $B_2 \subset \Gamma$ is a 2-subgroup whose image in $\Gamma/\Gamma_0 \simeq D_8$ is noncyclic, then $C_{\Gamma_0}(B_2) = T$.

Proof. (a) Replacing y by y^2 if necessary (which does not change E_y and enlarges the centralizer) we may assume η has order 2 modulo $F^\times$, so $s := \mathrm{Ad}(\eta)$ is an involutive automorphism of the pro-3 group Q. For $x \in Q$ put $b := (x^{-1}s(x))^{-1/2}$ (square roots are unique in a pro-3 group) and $a := xb^{-1}$; one checks $s(a) = a$, $s(b) = b^{-1}$ and $x^{-1}s(x) = b^{-2}$. If $[y,x] \in N$ then $b^{-2} \in \bar{N}$ (the image of N in Q, an s-stable closed normal subgroup), hence $b \in \bar{N}$ and $x \in Q^s\bar{N}$. Finally $Q^s = \{x : \eta x\eta^{-1} \in \pm xF^\times\}$; the sign $-$ would give a nontrivial 2-torsion element of Q, so $Q^s = (E_y^\times \cap F^\times K^1)/F^\times$, and x lifts into $\widetilde{E_y^\times}N\widetilde{Z}$.

(b) The (noncyclic) image of B_2 in D_8 contains the central involution $\bar{a}^2$ (the class of i) and a reflection. By (a) applied to the class of i, an element $b \in C_{\Gamma_0}(B_2)$ is the image of $\tilde{z}u$ with u a lift of $u_0 \in U^{(1)}_{E_0}$. A reflection η_2 acts on E_0 by σ and commutes with b in Γ, so $\sigma(u_0)u_0^{-1} \in \bar{N}$, i.e. $u_0^{-2} \in \bar{N}F^\times$, hence $u_0 \in \bar{N}$ and $b \in T$. □

8.3 The norm of Φ

Lemma 8.4 ([**P**])**.** *With respect to the probability Haar measures, $\langle\phi,\phi\rangle_{\widetilde{H}} = 1$ and, for $m \geq 1$,*

$$\langle \Phi|_{\widetilde{U}_m}, \Phi|_{\widetilde{U}_m}\rangle_{\widetilde{U}_m} = [Q_{m'} : Q_m], \qquad m' = \begin{cases} 1, & m \leq 3, \\ 2, & 4 \leq m \leq 6, \\ m-4, & m \geq 7. \end{cases}$$

In particular the norms for $m = 1, 3, 4, 6$ are 1, 27, 27, 729.

Proof. Weyl integration on the compact group H/Z reads $\int_{H/Z} f\,d\mu = \sum_E \frac{c_E}{2} \int_{E^\times/F^\times} \Delta(t)^2 f(t)\,dt$ for class functions f, with constants c_E fixed by $f \equiv 1$. On each torus the cube map $s \mapsto s^3$ is an injective endomorphism of $E^\times/F^\times$ with image the good elements, of index 9; it pushes Haar measure forward to $\frac{1}{9}$ of Haar measure on the image. Substituting $t = s^3$ and using $|\phi|^2(s^3) = 9R(s)^2$ and $R(s) = \Delta(s)/\Delta(s^3)$ gives $\int_{E^\times/F^\times} \Delta(t)^2|\phi|^2(t)\mathbf{1}_U(t)\,dt = \int \Delta(s)^2\mathbf{1}_U(s^3)\,ds$. For $U = H$ this is unitarity of the row. For $U = U_m$, note that $s^3 \in U_m$ iff $s \in U_{m'}$ by Lemma 7.1(c)–(e) and (a), so $\int_{U_m/Z} |\phi|^2 d\mu = \mu(U_{m'}/Z)$, and the normalized inner product on $\widetilde{U}_m$ is $\mu(U_{m'})/\mu(U_m) = [Q_{m'} : Q_m]$. □

8.4 The reduction theorem

Theorem 8.5 ([**P**] Reduction to the Sylow 3-subgroup)**.** *The following are equivalent.*

(i) (Target A) $L_D\mathbf{1}$ is the character of a genuine irreducible representation of $\widetilde{H}$ with central character $\widetilde{\theta}$.

(ii) $L_D\mathbf{1}$ is a virtual character of $\widetilde{H}$.

(iii) $\Phi = L_D\mathbf{1}|_{\widetilde{U}_1}$ is a virtual character of $\widetilde{U}_1$.

(iv) Φ is the character of an irreducible representation of $\Gamma_0 = \widetilde{U}_1/N$, necessarily of dimension 27.

Moreover, in that case the representation in (i) has dimension 27, its restriction to $\widetilde{U}_1$ is irreducible, and it is unique.

Proof. (i)⇒(ii)⇒(iii) are clear, and (iii)⇔(iv) by Lemma 8.4 ($\langle\Phi,\Phi\rangle = 1$, $\Phi(1) = 27 > 0$). Assume (iii). By Brauer's characterization of characters it suffices to show that $\phi|_B$ is a virtual character for every elementary subgroup $B = B_2 \times B_3 \subset \Gamma$ (B_ℓ an ℓ-group, one of them cyclic). If $B_2 = 1$, then $B \subset \Gamma_0$ and $\phi|_B = \Phi|_B$. If $B_2 = \langle y\rangle$ is cyclic and nontrivial, then $y \in T_{E_y}$ and $B_3 \subset C_{\Gamma_0}(y) \subset T_{E_y}$ by Lemma 8.3(a), so $B \subset T_{E_y}$ and Lemma 8.1 applies. If B_2 is noncyclic, then $B_3 \subset T$ by Lemma 8.3(b), so after conjugation $B \subset \Gamma_{P_0}$ and Lemma 8.2 applies. Hence ϕ is a virtual character of Γ; by unitarity (Lemma 8.4) and $\phi(1) = 27 > 0$ it is an irreducible character, which is (i). The restriction of that representation to $\widetilde{U}_1$ has character Φ, of norm one, hence is irreducible; a character determines the representation. □

Remark 8.6 (Independence of the choice of $\widetilde{\theta}$)**.** Two compatible choices $\widetilde{\theta}, \widetilde{\theta}'$ for $\rho = \mathbf{1}$ differ by a character θ_0 of $F^\times/F^{\times 3}$. Twisting a representation of $\widetilde{H}$ by $(\theta_0 \circ \mathrm{Nrd} \circ p)^{-1}$ changes its central character by $\theta_0(z)^{-2} = \theta_0(z)$ and multiplies its character at $\tilde{z}t^*$ by $\theta_0(\mathrm{Nrd}(zt^3))^{-1} = \theta_0(z)$; this is exactly the relation between $L_D\mathbf{1}$ computed with $\widetilde{\theta}$ and with $\widetilde{\theta}\theta_0$. So Target A does not depend on the choice.

9 The finite model and the defect function

Throughout this section we assume the following hypothesis, which is necessary for Target A (Lemma 9.2) and will be proved in Section 10.

Definition 9.1. Let $\psi_6 : \widetilde{U}_6 \to \mu_3$ be the function $\psi_6(\zeta \tilde{z}\, t^*) := \zeta \widetilde{\theta}(\tilde{z})$ on star lifts (all elements of $\widetilde{U}_6$ are star lifts, Lemma 7.1(d)). **Hypothesis (H1)**: ψ_6 is a (continuous) character of $\widetilde{U}_6$. Under (H1) put $N_6 := \ker \psi_6$, an open subgroup of $\widetilde{U}_6$, normal in $\widetilde{H}$ because the star section is conjugation equivariant and $\widetilde{\theta}$ is central.

Lemma 9.2 ([**P**])**.** *If $L_D\mathbf{1} = \Theta_\tau$ with τ irreducible, then $\tau|_{\widetilde{U}_6} = 27\,\psi_6$; in particular (H1) holds.*

Proof. By Lemma 7.3, $|\Theta_\tau| = 27 = \dim \tau$ on $\widetilde{U}_6$, so $\widetilde{U}_6$ acts by scalars, by ψ_6. □

Under (H1) define

$$\bar{\Gamma} := \widetilde{U}_1/N_6 \text{ (order } 3^9), \qquad \widetilde{V} := \widetilde{U}_3/N_6, \qquad \mu_3 \simeq \widetilde{U}_6/N_6 \subset \widetilde{V} \to V = Q_3/Q_6 \to 0.$$

$\widetilde{V}$ is a central extension of the elementary abelian group V by μ_3; let $\omega : V \times V \to \mu_3$ be its commutator form. By Lemma 7.3, Φ factors through $\bar{\Gamma}$, with $\bar{\Gamma}_6 := \widetilde{U}_6/N_6 = \mu_3$ carrying $\Phi = 27\,\mathrm{id}$. The elements $\bar{\sigma}(g)$ ($g \in G_3$) are the images of the star lifts; by Lemma 7.1(f) and the next lemma they depend only on the class of g in $\bar{G}_3 \subset V$.

Lemma 9.3 ([**P**] Three consequences of (H1))**.** *Assume (H1).*

(a) For $a \in H$ and $b \in U_6$, $\psi_6([\tilde{a}, \tilde{b}]) = 1$. Hence for $i + j \geq 6$ the map $\kappa_{i,j}(a, b) := \psi_6([\tilde{a}, \tilde{b}])$ on $K^i \times K^j$ is bimultiplicative, alternating, conjugation invariant, and trivial if a or b lies in U_6; $\kappa_{1,5}$ factors through $(Q_1/Q_2) \times (Q_5/Q_6) \simeq \mathbb{F}_9 \times \mathbb{F}_9$.

(b) $\kappa_{1,5}(a, b) = \zeta^{\beta \operatorname{Tr}(a\bar{b})}$ for a fixed sign $\beta \in \{\pm 1\}$.

(c) (H3) $\omega = 0$: $\widetilde{V}$ is abelian, hence $\widetilde{V} \simeq \mu_3 \times V$.

(d) (H2) For $g \in G_3$ and $u \in Q_6$, $\bar{\sigma}(gu) = \bar{\sigma}(g)$.

Proof. (a) $\ker \psi_6$ is normal in $\widetilde{H}$ and $\widetilde{U}_6/\ker \psi_6$ is central in $\widetilde{H}/\ker \psi_6$ (as ψ_6 is conjugation invariant), so $[\tilde{a}, \tilde{b}] \in \ker \psi_6$ for $b \in U_6$. Bimultiplicativity follows from $[xy, z] = x[y, z]x^{-1}[x, z]$. A character of Q trivial on cubes and commutators factors through Q/Q_2, since $[Q, Q]Q^3 \supseteq Q_2$ (level by level: at even levels the commutators $[1 + a\pi, 1 + b\pi^{k-1}]$ produce the trace-zero line and U_F the rest; at odd levels they produce all of $\mathbb{F}_9$).

(b) Write $\kappa_{1,5}(a, b) = \zeta^{\operatorname{Tr}(\alpha ab) + \operatorname{Tr}(\beta a\bar{b})}$ with $\alpha, \beta \in \mathbb{F}_9$ (these span the bilinear forms). Conjugation by $\xi \in \mathcal{O}_{E_0}^\times$ multiplies both a and b by $\lambda = \xi/\bar{\xi} \in \mu_4$, forcing $\alpha = 0$; conjugation by j applies Frobenius to both, forcing $\beta \in \mathbb{F}_3$. Finally, in a ramified torus the depth-one and depth-five classes span the two-dimensional quotient A_E/A_E^3 on which the covered-torus pairing (10) is nondegenerate (Part I, Proposition 3.5), while the leading coefficients of a depth-one and a depth-five element of the same torus are $\mathbb{F}_3^\times$-proportional; hence $\kappa_{1,5}(a, ra) \neq 1$ for $a \neq 0$, $r \in \mathbb{F}_3^\times$, i.e. $\beta \neq 0$.

(c) Let t have depth one and $y \in K^4$, $c := [\tilde{t}, \tilde{y}] \in \tilde{K}^5$, $d := [c^{-1}, \tilde{t}] \in \tilde{K}^6$, $e := [d^{-1}, \tilde{t}]$. Then $[\tilde{t}^3, \tilde{y}] = c^{t^2} c^t c$ and, expanding with $c^t = cd$, $c^{t^2} = cd^2 e$, one gets $[\tilde{t}^3, \tilde{y}] = c^3 \cdot Y$ with Y a product of d^3, e and commutators each involving d or e. As c^3 is a star lift, $d, e \in \widetilde{U}_6$ and (a) holds, $\psi_6([\tilde{t}^3, \tilde{y}]) = 1$. Thus $\omega(\bar{G}_3, V_4) = 0$ with $V_4 = Q_4/Q_6$, hence $\omega(V, V_4) = 0$ since $\bar{G}_3$ spans V (Lemma 7.1(g)). So ω is induced from an alternating form on $V/V_4 = Q_3/Q_4 \simeq \mathbb{F}_9$, a two-dimensional $\mathbb{F}_3$-space, on which conjugation by j acts by Frobenius, of determinant -1; invariance forces $\omega = 0$. So $\widetilde{V}$ is

abelian. It also has exponent 3: for $v \in K^3$ the cube of any lift is the star lift of $v^3 \in K^9 \subset U_6$, on which $\psi_6 = 1$, and for $z \in F^\times$ the cube of $\tilde{z}$ is the star lift of z^3, on which $\psi_6 = \theta_0(z)^3 = 1$; so every element of $\widetilde{V} = \widetilde{U}_3/N_6$ cubes into N_6. An abelian group of exponent 3 is an $\mathbb{F}_3$-vector space, so the subgroup μ_3 has a complement. (Abelianness alone would not suffice: $C_9 \to C_3$ is a nonsplit abelian extension with genuine characters.)

(d) Let $g = t^3$, t of depth one with leading coefficient a, and $gu = t'^3$ with $t' = t(1+w)$, $w \in \mathbb{F}_3 a\pi^3 + \mathfrak{P}^4$ (proof of Lemma 7.1(f)). In $\bar{\Gamma}$ put $x = \tilde{t}$, $y = \widetilde{1+w}$, $c = [y,x]_L = yxy^{-1}x^{-1}$. The ring computation in Lemma 7.1(f) shows $[1+w, t] \in K^5$ with level-five coefficient in $\mathbb{F}_3\, ia$, so $c \in \bar{\Gamma}_5$. Since $\bar{\Gamma}_2$ has exponent 3 (cubes of $\widetilde{U}_2$ are star lifts), $c^3 = 1$; $[c,y] \in \bar{\Gamma}_8 = \mu_3$ equals ω-type pairing $\kappa_{5,3}$, trivial by (c); and $d := c^{-1}xcx^{-1} \in \bar{\Gamma}_6 = \mu_3$ is central. A direct expansion then gives $(xy)^3 = d^{-1}x^3y^3$, and $y^3 = 1$ in $\bar{\Gamma}$ because $(1+w)^3 \in Q_9$. Hence $\bar{\sigma}(gu)\bar{\sigma}(g)^{-1} = d^{-1} = \kappa_{1,5}(a,\, s\, ia)^{\pm 1}$ for some $s \in \mathbb{F}_3$, and $\mathrm{Tr}(a\,\overline{s\, i\, a}) = -s\,\mathrm{N}(a)\,\mathrm{Tr}(i) = 0$. □

9.1 The 27 characters and the defect function

Assume (H1). Fix a complement V_0 to μ_3 in $\widetilde{V}$; genuine characters of $\widetilde{V}$ are then $\mu_h(\zeta, v) = \zeta \cdot \zeta^{h(v)}$, $h \in V^\vee$. The star section gives a canonical lift $\bar{\sigma}(v) \in \widetilde{V}$ for each $v \in \bar{G}_3$, and we define the *defect function* $e : \bar{G}_3 \to \mathbb{F}_3$ by $\bar{\sigma}(v) = (\zeta^{e(v)}, v)$; it satisfies $e(-v) = -e(v)$ (star is a homomorphism on each torus line). The group $\mathrm{Heis} := Q_1/Q_3$ (extraspecial of order 27 and exponent 3) acts on $\widetilde{V}$ by conjugation, fixing μ_3; relative to V_0 the action is $(\zeta, v) \mapsto (\zeta\zeta^{\beta_s(v)}, sv)$ with $\beta_s \in V^\vee$, and equivariance of the star section gives

$$e(sv) - e(v) = \beta_s(v) \qquad (v \in \bar{G}_3,\ s \in \mathrm{Heis}). \tag{24}$$

Proposition 9.4 ([**P**] What Target A forces)**.** *Suppose $L_D\mathbf{1} = \Theta_\tau$. Then, with $\Lambda \subset V^\vee$ the set of h with μ_h a constituent of $\tau|_{\widetilde{V}}$:*

(a) $\tau|_{\widetilde{V}}$ is multiplicity free, $|\Lambda| = 27$, and Heis *acts simply transitively on Λ (by $h \mapsto (h - \beta_s) \circ s^{-1}$);*

(b) for $v \in V \setminus (\bar{G}_3 \cup \{0\})$ the values $h(v)$, $h \in \Lambda$, are equidistributed in $\mathbb{F}_3$;

(c) for $v \in \bar{G}_3$ the value $-e(v)$ is taken 15 times and the other two values 6 times each;

(d) equivalently, for every $h \in V^\vee$ the number $N_e(h) := \#\{\text{lines } \ell \subset \bar{G}_3 \cup \{0\} :\ h|_\ell = -e|_\ell\}$ is 20 if $h \in \Lambda$ and 11 otherwise.

Conversely, if (H1) holds and e satisfies (d) for some 27-set Λ on which Heis *acts simply transitively, then $\tau_1 := \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1} \lambda$, $\lambda = \mu_h$ for any $h \in \Lambda$, is irreducible with character Φ, and Target A holds by Theorem 8.5.*

Proof. By Lemma 8.4, $\langle \Phi|_{\widetilde{U}_3}, \Phi|_{\widetilde{U}_3} \rangle = 27 = \Phi(1)$, which forces multiplicity one with 27 constituents; Clifford theory for the normal subgroup $\widetilde{U}_3$ of $\widetilde{U}_1$, with $[\widetilde{U}_1 : \widetilde{U}_3] = 27$, gives (a). Evaluating $\sum_{h \in \Lambda} \mu_h = \Phi|_{\widetilde{V}}$ at $(1, v)$ and at $\bar{\sigma}(v)$ gives (b) and (c), using (21) and (H2); (d) is the Fourier dual statement (each line $\{v, -v\}$ contributes 2 or -1 to $\sum_{v \in \bar{G}_3} \zeta^{h(v)+e(v)} = 3N_e(h) - 36 \in \{24, -3\}$). The converse is Mackey irreducibility and the Frobenius character formula, whose Fourier transform is (b)–(c), together with the vanishing of Φ and of the induced character off $\widetilde{U}_3$. □

9.2 Symmetry pins down the defect function

The group $D^\times$ acts on everything by conjugation through $G_{\mathrm{sym}} := D^\times/F^\times K^3$, of order 216, an extension of $D_8 = \langle \mathrm{Ad}(1+i), \mathrm{Ad}(j)\rangle$ by Heis. Its action on V is recorded in Appendix A; one checks $V^{D_8} = 0$ and $(V^\vee)^{D_8} = 0$. Since $|D_8|$ is prime to 3, there is a D_8-stable complement V_0, and it is unique. With respect to it, e is D_8-invariant. The D_8-orbits on $\bar{G}_3$ have sizes $4, 4, 8, 8, 8, 8, 8, 8, 8, 8$; six of them contain $-v$ together with v, forcing $e = 0$ there, and the other four form two pairs $\{O, -O\}$. Hence e is determined by two parameters in $\mathbb{F}_3$.

Proposition 9.5 ([**P**^{c}] Uniqueness of the design)**.** *Among the 3^{10} D_8-invariant functions e on $\bar{G}_3$, exactly two are odd and satisfy condition (d) of Proposition 9.4, namely e_* and $-e_*$, where e_* is tabulated in Appendix A; it vanishes on 20 of the 36 lines and takes the values 1 and 2 on 8 lines each. Among the nine odd D_8-invariant e, exactly three admit linear β_s as in (24) for the generators $s = \mathrm{Ad}(1+j), \mathrm{Ad}(1+ij)$ of* Heis*: $e = 0$ and $e = \pm e_*$. For $e = \pm e_*$ the resulting* Heis*-action on $V^\vee$ is simply transitive on $\Lambda_\pm := \{h : N_{\pm e_*}(h) = 20\}$, $0 \in \Lambda_\pm$, and (b)–(c) hold. For $e = 0$ one has $\beta_s = 0$ for all s.*

Proof. Exact enumeration over $\mathbb{F}_3$ (Appendix B, scripts 3, 4, 6): the D_8-orbits, the counts $N_e(h)$ for all 243 covectors and all candidates, the linear extension problem (24), the orbit of 0 under the affine action, and the Fourier identities. The distribution of $N_0(h)$ is $\{9^{80}, 12^{90}, 15^{72}, 36^1\}$, that of $N_{\pm e_*}(h)$ is $\{11^{216}, 20^{27}\}$. □

Theorem 9.6 ([**P**^{c}] Target A is equivalent to (H1))**.** *$L_D\mathbf{1}$ is a genuine irreducible character if and only if (H1) holds. In that case the defect function is $\pm e_*$, and $\tau|_{\widetilde{U}_1} \simeq \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1} \lambda_0$ where λ_0 is the genuine character of $\widetilde{U}_3$ trivial on N_6 and on the unique D_8-stable complement V_0; explicitly*

$$\lambda_0(y) = \Big(\prod_{d \in D_8} \tilde{d}\, y\, \tilde{d}^{-1} \Big)^{-1} \in \mu_3 \qquad (y \in \widetilde{U}_3), \tag{25}$$

the product taken in the abelian group $\widetilde{V}$ (it lies in $\widetilde{V}^{D_8} = \mu_3$).

Proof. Necessity is Lemma 9.2. Assume (H1). By Lemma 9.3, (H2) and (H3) hold, so the defect function e is defined, is odd and D_8-invariant for the D_8-stable complement, and satisfies (24); by Proposition 9.5, $e \in \{0, e_*, -e_*\}$. If $e = 0$, then $\beta_s = 0$ for $s = \mathrm{Ad}(1+j)$; but for the depth-five vector $v = e_{B_{\mathrm{re}}}$ of the torus $F(j)$, fixed by s, $\zeta^{\beta_s(v)}$ is the commutator of lifts, i.e. $\kappa_{1,5}(1,1) = \zeta^{2\beta} \neq 1$ by Lemma 9.3(b). Hence $e = \pm e_*$, and the converse part of Proposition 9.4 applies with $\Lambda = \Lambda_\pm \ni 0$; the character μ_0 is λ_0, and (25) is the formula for the projection $\widetilde{V} = \mu_3 \times V_0 \to \mu_3$, since $\prod_d d \cdot x = \lambda_0(x)^8 = \lambda_0(x)^{-1}$ for $x \in \widetilde{V}$. Uniqueness of V_0: complements form a torsor under $V^\vee$, and $(V^\vee)^{D_8} = 0$. □

Remark 9.7. The two signs correspond to the two identifications of μ_3 with $\mathbb{Z}/3$ (equivalently, to replacing the cover by its opposite); the statement of Theorem 9.6 is symmetric. The Heis-cocycle β attached to e_* restricted to $V_5 = Q_5/Q_6$ reproduces $\kappa_{1,5}$ with $\beta = -1$, an independent consistency check between the design computation and the torus data.

10 The cover in the Kazhdan–Patterson model

It remains to prove (H1). We do this in an explicit model of the cover. By Part I, Definition 2.2 and Proposition 2.3(a), the cover of $D^\times$ is the restriction, along

$$\iota : D^\times \hookrightarrow \mathrm{GL}_2(E_0), \qquad x + yj \longmapsto \begin{pmatrix} x & \varpi y \\ \sigma(y) & \sigma(x) \end{pmatrix}, \tag{26}$$

of the parameter-c_v Kazhdan–Patterson cover of $\mathrm{GL}_2(E_0)$, whose cocycle is $\sigma_{E_0}(g,h)\,(\det g, \det h)_{E_0}^{c_v}$ with σ_{E_0} the parameter-0 cocycle. We use the Kubota-type representative

$$\sigma_{E_0}(g,h) = \Big(\frac{x(gh)}{x(g)},\ \frac{x(gh)}{x(h)\det g}\Big)_{E_0}, \qquad x\begin{pmatrix} a & b \\ c & d\end{pmatrix} = \begin{cases} c, & c \neq 0, \\ d, & c = 0, \end{cases} \tag{27}$$

and write $\beta(g,h) \in \mathbb{Z}/3$ for the exponent of $\sigma_{E_0}(g,h)(\det g, \det h)_{E_0}^2$ (before the relabeling $\zeta \mapsto \zeta^{r'}$, which multiplies all exponents by 2 and is irrelevant for everything below). Elements of the cover are pairs (g,e) with $(g,e)(h,f) = (gh, e+f+\beta(g,h))$.

Lemma 10.1 ([**E**]/[**P**] The model is the cover)**.** *(a)* (27) *is a 2-cocycle on* $\mathrm{GL}_2(E_0)$ *whose restriction to the diagonal torus is* $(a_1,b_2)_{E_0}$.

(b) On scalars the twisted cocycle is trivial, so $z \mapsto (z,0)$ *is a homomorphic section over* $F^\times$*; and for* $z \in F^\times$, $g \in \mathrm{GL}_2(E_0)$, $\beta(z,g) = (z,\, x(g)\det g)_{E_0}$.

(c) For $x, y \in E_0^\times$*, the commutator of* $(\iota x, 0)$ *and* $(\iota y, 0)$ *has exponent* $(\mathrm{N}\,x, \mathrm{N}\,y)_F + (x,y)_{E_0}$*, in accordance with Part I, Proposition 3.1 (before relabeling,* $c = 1$*).*

(d) The extension so defined is isomorphic, as a central extension of $D^\times$ *by* μ_3*, to the cover of Part I; any two such isomorphisms differ by* $\chi \circ \mathrm{Nrd}$ *for a character* χ *of* $F^\times$ *with values in* μ_3*, which does not affect star lifts, good elements, or* $L_D\mathbf{1}$.

Proof. (a) is Kubota's cocycle for GL_2: for SL_2 it is [Kub67, Thm. 2] and [Kub, Ch. 1], and the extension to GL_2 by the factor $\det g$ in the second entry is the case $r = 2$, $c = 0$ of the Kazhdan–Patterson cocycle [KP1, §0.1], with the n-th power Hilbert symbol for the n-fold cover (the same formula serves in Section 13); it is the one external input of this section and is labeled [**E**]. (The twist $(\det g, \det h)^c$ is bimultiplicative, hence a cocycle, so the twisted formula is a cocycle as well.) As a safeguard against transcription errors, the cocycle identity was also checked on 250 random triples and the diagonal restriction on 100 random pairs by exact computation (script 7); this check is corroboration, not part of the proof. (b) With $x(zI) = z$, $x(zg) = zx(g)$, $\det(zI) = z^2$ one gets $\sigma(zI,g) = (x(g), z^{-1}) = (z, x(g))$ and the twist contributes $(z^2, \det g)^2 = (z, \det g)$; for $g = wI$ the two terms cancel, since $(z,w)_{E_0} = (z,w)_F^2$. (c) is a direct evaluation; it was also checked on 200 random pairs. (d) Steps 1–3 of the proof of Part I, Proposition 2.3(b), apply verbatim to two cocycles on $\mathrm{GL}_2(E_0)$ with the same diagonal restriction; the last assertion is Part I, Remark 8.1 (the star map $x^* = \tilde{x}^{\,3}$ absorbs $\chi \circ \det$). □

The Hilbert symbols. The cubic Hilbert symbols of F and of E_0 enter only through their classes. By Moore's theorem (cited in Part I, Proposition 2.3), a continuous Steinberg symbol $F^\times \times F^\times \to \mu_3$ is a power of the Hilbert symbol. We therefore computed the alternating bilinear forms on $F^\times/F^{\times 3} \simeq \mathbb{F}_3^4$ and $E_0^\times/E_0^{\times 3} \simeq \mathbb{F}_3^6$ annihilated by all Steinberg relations $(a, 1-a) = 1$ (exact linear algebra over $\mathbb{F}_3$ on several thousand relations, script 7); in both cases the solution space is one-dimensional and nondegenerate, so each symbol is determined up to the common sign, and the relative sign is fixed by norm compatibility $(a,b)_{E_0} = (a, \mathrm{N}\,b)_F$, which the computed forms satisfy. Classes modulo cubes are computed from the filtrations $U^{(k)}$ by explicit discrete logarithms. The hand arguments below use only the following standard facts about the cubic symbol of E_0, for which $e(E_0/\mathbb{Q}_3) = 2$ and $pe/(p-1) = 3$: bimultiplicativity and antisymmetry; the Steinberg relation $(u, 1-u) = 1$; Galois invariance $(\sigma u, \sigma v) = (u,v)$; triviality on cubes, in particular on -1,

on μ_8 and on $U_{E_0}^{(4)} \subset E_0^{\times 3}$; and the depth bound $(U_{E_0}^{(i)}, U_{E_0}^{(j)}) = 1$ for $i + j \geq 4$ (see [FV, Ch. IV] or [Ser, Ch. XIV]). A consequence used repeatedly: for $m \in U_{E_0}^{(3)}$ and any $u \in E_0^{\times}$,

$$(u, m)_{E_0} = (\varpi, m)_{E_0}^{v_{E_0}(u)}, \tag{28}$$

because $u = \varpi^{v(u)} \xi u_1$ with $\xi \in \mu_8$ and $u_1 \in U_{E_0}^{(1)}$. The computed tables are consistent with these facts (for instance $(1 + \varpi, 1 + \varpi^3) = (1 + \varpi^2, 1 + \varpi^2 i) = 1$ while $(1 + \varpi, 1 + \varpi^2) \neq 1$). With $\varpi,\ 1 + \varpi,\ 1 + \varpi^2,\ 1 + \varpi^3$ as basis of $F^{\times}/F^{\times 3}$ the form is $(\varpi, 1 + \varpi^3) = (1 + \varpi, 1 + \varpi^2) = \zeta$ and all other basis pairings trivial; the 6×6 matrix for E_0 is in Appendix A.

10.1 The cube identity and the star section on K^6

Proposition 10.2 ([**P**] Cubes of the standard section)**.** *Let $w = x + yj \in D^{\times}$ with $y \neq 0$, $T := \mathrm{Tr}_{E_0/F}(x) \neq 0$ and $A := x^2 + x\sigma(x) + \sigma(x)^2 + \varpi\,\mathrm{N}(y) \neq 0$. Then $(w, 0)^3 = (w^3, 0)$ in the cover. The hypotheses hold for every $w \in K^1 \setminus E_0$; for $w \in E_0^{\times}$ the conclusion holds as well.*

Proof. Put $n = \mathrm{Nrd}(w) = \mathrm{N}(x) - \varpi\,\mathrm{N}(y)$. From $w^2 = (x^2 + \varpi\,\mathrm{N}\,y) + yTj$ and $w^3 = \cdots + yA\,j$ one reads off $x(\iota w) = \sigma(y)$, $x(\iota w^2) = \sigma(y)T$, $x(\iota w^3) = \sigma(y)A$, with $T, A \in F$. Writing h_E, h_F for the exponents of the symbols and using $h_E(a, b) = 2h_F(a, b)$ for $a, b \in F$:

$$\beta(w, w) = h_E(T, T/n) = h_F(T, n), \qquad \beta(w^2, w) = h_E(A/T, A/n^2) = h_F(nT, A) + h_F(T, n),$$

the twists vanishing because $h_E(n, n) = h_E(n^2, n) = 0$. Hence the exponent of $(w, 0)^3$ is $h_F(n, T) + h_F(nT, A)$. Now $A = T^2 - n$; with $u := n/T^2$,

$$h_F(nT, T^2 - n) = h_F(nT, T^2) + h_F(nT, 1 - u) = 2h_F(n, T) + h_F(uT^3, 1 - u) = 2h_F(n, T),$$

by the Steinberg relation $h_F(u, 1 - u) = 0$. So the exponent is $3h_F(n, T) = 0$. For $w \in K^1$, $x \equiv 1 \bmod \varpi$ gives $T \equiv 2$ and $v_F(x^2 + x\sigma x + \sigma x^2) = 2$ exactly, so $A \neq 0$. For $w = x \in E_0^{\times}$, $x(\iota w^k) = \sigma(x)^k$ and $\det = \mathrm{N}(x)^k$, and the same bookkeeping gives $\sigma(w, w) + \sigma(w^2, w) = -h_E(\sigma x, x) + h_E(\sigma x, x) = 0$. □

Corollary 10.3 ([**P**] The star section on K^6)**.** *(a) For every cube $g = t^3$ with $t \in K^1$, the star lift is $\sigma(g) = (g, 0)$.*

(b) Every $X \in K^6$ can be written $X = zv^3$ with $z \in U_F^{(1)}$ and $v \in K^2$ (Lemma 7.1(d)); z is determined modulo $F^{\times 3}$, and $z \equiv \mathrm{Nrd}(X)^{-1} \pmod{F^{\times 3}}$.

(c) The star lift of $X = x_X + y_X j \in K^6$, that is, the element $\ell(X) := \tilde{z}\,\tilde{v}^3$ with $\tilde{z} = (z, 0)$, is

$$\ell(X) = (X, \kappa(X)), \qquad \kappa(X) = \begin{cases} h_E(y_X, \mathrm{Nrd}\,X), & y_X \neq 0, \\ 0, & y_X = 0. \end{cases} \tag{29}$$

Equivalently, $\kappa(X) = c(X)\, v_{E_0}(y_X)$, where $c(X) \in \mathbb{F}_3$ is the real part of the level-six digit of X (the digit of $(x_X - 1)/\varpi^3$).

(d) (H1) is equivalent to the identity

$$\kappa(XX') = \kappa(X) + \kappa(X') + \beta(X, X') \qquad (X, X' \in K^6), \tag{30}$$

and when it holds, ψ_6 is given on $\tilde{K}^6$ by

$$\psi_6(X,e) = \zeta^{\,e-\kappa(X)}\,\theta_0(\mathrm{Nrd}\,X)^{-1}, \tag{31}$$

where $\widetilde{\theta}(\zeta,z) = \zeta\theta_0(z)$ on $\widetilde{Z} = \mu_3 \times F^\times$, the splitting being $z \mapsto (z,0)$ of Lemma 10.1(b), and θ_0 is any character with $\theta_0^3 = 1$.

Proof. (a) is Proposition 10.2. (b) Existence is Lemma 7.1(d) ($\mathrm{Cubes}(Q_2) = Q_6$). Taking reduced norms, $\mathrm{Nrd}(X) = z^2\,\mathrm{Nrd}(v)^3$, so $z^2 \equiv \mathrm{Nrd}\,X$ modulo $F^{\times 3}$; if also $X = z'v'^3$ then $(z/z')^2 \in F^{\times 3}$, hence $z/z' \in F^{\times 3}$ because $F^\times/F^{\times 3}$ has exponent 3. Finally $z \equiv z^4 \equiv \mathrm{Nrd}(X)^2 \equiv \mathrm{Nrd}(X)^{-1}$ modulo cubes. (c) $\ell(X) = (z,0)(v,0)^3 = (z,0)(v^3,0) = (X, \beta(z,v^3))$ by (a), and $\beta(z,v^3) = h_E(z, x(\iota v^3)\det \iota v^3)$ by Lemma 10.1(b). If $y_X \neq 0$ then $x(\iota v^3) = \sigma(y_X/z)$ and $\det \iota v^3 = \mathrm{Nrd}(v)^3$, so $\beta(z,v^3) = h_E(z,\sigma(y_X)) - h_E(z,z) = h_E(z,y_X)$ by Galois invariance and $h_E(z,z) = h_E(z,-1) = 0$; by (b) this is $-h_E(\mathrm{Nrd}\,X, y_X) = h_E(y_X, \mathrm{Nrd}\,X)$, which does not depend on the choice of z because $h_E(F^{\times 3},\cdot) = 0$. If $y_X = 0$ then $x(\iota v^3) = \sigma(x_X/z)$, $\beta(z,v^3) = h_E(z,x_X) = h_E(x_X, \mathrm{Nrd}\,X) = h_E(x_X, x_X\sigma x_X) = h_E(x_X, \sigma x_X) = 0$, since $x_X \in U_{E_0}^{(3)}$ and $3+3 \geq 4$. The digit formula follows from (28), from $h_E(\varpi, \mathrm{Nrd}\,X) = h_E(\varpi, \mathrm{N}\,x_X) = 2h_E(\varpi, x_X) = -h_E(\varpi, x_X)$ (as $\mathrm{Nrd}\,X \equiv \mathrm{N}\,x_X$ modulo $U_F^{(7)} \subset F^{\times 3}$ and $h_E(\varpi, \sigma x_X) = h_E(\varpi, x_X)$), and from $h_E(1+\varpi^3, \varpi) = 1$, $h_E(1+\varpi^3 i, \varpi) = 0$ (Appendix A); it is recorded because the tables and scripts use it. (d) $\ell(X)\ell(X') = (XX', \kappa(X) + \kappa(X') + \beta(X,X'))$, so (30) says exactly that ℓ is a homomorphism on K^6. If it is, then ℓ is a homomorphism on $U_6 = F^\times K^6$, because $\ell(zX) = (z,0)\ell(X)$ with $(z,0)$ central and $z \mapsto (z,0)$ a homomorphism (Lemma 10.1(b)); and then $\widetilde{U}_6 = \mu_3 \times \ell(U_6)$ and ψ_6 is the character which is $\zeta \mapsto \zeta$ on μ_3 and $zv^3 \mapsto \theta_0(z)$ on $\ell(U_6)$. The latter is well defined and multiplicative on U_6: if $zv^3 \cdot z'v'^3 = z''v''^3$ then $z'' \in zz'F^{\times 3}$ by the norm argument of (b), and $\theta_0^3 = 1$. Since $(X,e) = \zeta^{e-\kappa(X)}\ell(X)$ and $\theta_0(z) = \theta_0(\mathrm{Nrd}\,X)^{-1}$ by (b), this is (31); the factor $\theta_0(\mathrm{Nrd}\,X)^{-1}$ is itself a character of K^6, so the choice of θ_0 affects neither (H1) nor ℓ. Conversely (H1) implies (30), since ψ_6 is then a character whose kernel meets each fiber of $\widetilde{U}_6 \to U_6$ in exactly one point, namely $\ell(u)$. □

Proposition 10.4 (**[P]** (H1) holds)**.** *The identity (30) holds for all $X, X' \in K^6$. Consequently ψ_6 is a character of $\widetilde{U}_6$, given on $\tilde{K}^6$ by (31).*

Proof. Write $X = a + bj$, $X' = a' + b'j$ with $a, a' \in U_{E_0}^{(3)}$, $b, b' \in \varpi^3\mathcal{O}_{E_0}$, and $n = \mathrm{Nrd}\,X$, $n' = \mathrm{Nrd}\,X'$, so that $n, n' \in U_F^{(3)}$ and $XX' = (aa' + \varpi b\sigma(b')) + (ab' + b\sigma(a'))j$. Three preliminary remarks. First, $n = \mathrm{N}(a) - \varpi\,\mathrm{N}(b) \equiv \mathrm{N}(a)$ modulo $U_F^{(7)} \subset E_0^{\times 3}$, and likewise $n' \equiv \mathrm{N}(a')$, so in every symbol n and n' may be replaced by $\mathrm{N}\,a$ and $\mathrm{N}\,a'$. Second, by (28) and Galois invariance, for every $u \in E_0^\times$,

$$h_E(u, \mathrm{N}\,a) = v(u)\,\lambda(a), \qquad \lambda(a) := h_E(\varpi, \mathrm{N}\,a) = 2h_E(\varpi, a) = h_E(a, \varpi), \tag{32}$$

and similarly for a'. Third, all symbols between two of $a, \sigma a, a', \sigma a', n, n'$ vanish ($3+3 \geq 4$), and the twist $2h_E(n,n')$ vanishes. We write v for v_{E_0}.

Main case: $b \neq 0$, $b' \neq 0$, $R \neq 0$, where

$$\tau := \sigma(b'/b), \qquad R := a' + \sigma(a)\tau,$$

so that $x(\iota X) = \sigma(b)$, $x(\iota X') = \sigma(b')$ and $x(\iota XX') = \sigma(ab' + b\sigma(a')) = \sigma(b)R$. Inserting into (27) and using $h_E(R,R) = h_E(R,-1) = 0$,

$$\beta(X,X') = h_E\Big(R, \frac{R}{\tau n}\Big) + 2h_E(n,n') = -h_E(R,\tau) - h_E(R,n). \tag{33}$$

On the other side, $y_{XX'} = ab' + b\sigma(a') = b\,\sigma(R)$, so by (29), Galois invariance and the first remark,

$$\kappa(XX') = h_E(b\sigma(R), nn') = h_E(b,n) + h_E(b,n') + h_E(R,n) + h_E(R,n').$$

Hence (30) is equivalent to $h_E(b,n') + h_E(R,n) + h_E(R,n') = h_E(b',n') - h_E(R,\tau) - h_E(R,n)$, i.e. (using $h_E(b'/b,n') = h_E(\sigma\tau,n') = h_E(\tau,n')$) to

$$h_E(R,\tau) = h_E(\tau,n') - h_E(R,n') + h_E(R,n) = v(\tau)\lambda(a') - v(R)(\lambda(a') - \lambda(a)), \tag{34}$$

the second equality by (32). To prove (34), factor $R = a'(1+\tau_2)$ with $\tau_2 := \sigma(a)\tau/a'$; then $\tau_2 \neq 0, -1$ and $\tau = \tau_2 \cdot a'/\sigma(a)$, so

$$h_E(R,\tau) = h_E(a',\tau) + h_E(1+\tau_2,\tau_2) + h_E(1+\tau_2,\, a'/\sigma(a)).$$

The first term is $v(\tau)\lambda(a')$ by (28) and (32). The second vanishes by the Steinberg relation ($h_E(1+\tau_2,\tau_2) = h_E(1-(-\tau_2),-\tau_2) + h_E(1+\tau_2,-1) = 0$). For the third, $a'/\sigma(a) \in U_{E_0}^{(3)}$ and $v(1+\tau_2) = v(R)$, so by (28) it equals $v(R)\big(h_E(\varpi,a') - h_E(\varpi,\sigma a)\big) = v(R)\big(h_E(\varpi,a') - h_E(\varpi,a)\big) = -v(R)(\lambda(a') - \lambda(a))$. This is (34). Note that no case distinction on $v(\tau)$ or on $v(1+\tau)$ was needed.

Case $b = 0$, $b' \neq 0$. Then $x(\iota X) = \sigma(a)$, $x(\iota XX') = \sigma(a)\sigma(b')$, and (27) gives $\beta = h_E(\sigma b', \sigma a/n) = h_E(\sigma b', 1/a) = -h_E(b',\sigma a)$. Also $\kappa(X) = 0$, $\kappa(X') = h_E(b',n')$ and $\kappa(XX') = h_E(ab', nn') = h_E(a,n) + h_E(a,n') + h_E(b',n) + h_E(b',n')$, with $h_E(a,n) = h_E(a,\sigma a) = 0$ and $h_E(a,n') = 0$ by the third remark. So (30) reduces to $h_E(b',n) + h_E(b',\sigma a) = 0$; by the first remark $n\sigma(a) \equiv a\sigma(a)^2$, so the left side is $h_E(b', a/\sigma a) + 3h_E(b',\sigma a) = v(b')\big(h_E(\varpi,a) - h_E(\varpi,\sigma a)\big) = 0$.

Case $b \neq 0$, $b' = 0$. Then $x(\iota X') = \sigma(a')$, $x(\iota XX') = \sigma(b)a'$, and $\beta = h_E(a', \sigma(b)a'/(\sigma(a')n)) = h_E(a',\sigma b) = v(b)\lambda(a')$, the other three symbols vanishing by the third remark. Also $\kappa(X') = 0$ and $\kappa(XX') = h_E(b\sigma(a'), nn') = h_E(b,n) + h_E(b,n') = \kappa(X) + v(b)\lambda(a')$. Both sides of (30) equal $\kappa(X) + v(b)\lambda(a')$.

Case $b = b' = 0$. All three κ vanish and $\beta = h_E(a,\sigma a') = 0$.

Case $b, b' \neq 0$, $R = 0$. Then $XX' \in E_0$, so $\kappa(XX') = 0$, and $\tau = -a'/\sigma(a)$, $b' = \sigma(\tau)b = -b\,\sigma(a')/a$. Now $x(\iota XX') = \sigma(aa' + \varpi b\sigma b')$ and $aa' + \varpi b\sigma(b') = aa' + \varpi\,\mathrm{N}(b)\tau = (a'/\sigma a)\,n$, so $x(\iota XX') = \sigma(a')n/a$ and

$$\beta = h_E\Big(\frac{\sigma(a')n}{a\,\sigma(b)},\ \frac{\sigma(a')}{a\,\sigma(b')}\Big) + 2h_E(n,n').$$

Expanding bimultiplicatively, discarding the symbols that vanish by the third remark, and using Galois invariance, (28) and $v(b') = v(b)$:

$$\begin{aligned}\beta &= -h_E(\sigma a', \sigma b') - h_E(n,\sigma b') + h_E(a,\sigma b') - h_E(\sigma b,\sigma a') + h_E(\sigma b, a) + h_E(\sigma b,\sigma b')\\ &= v(b)\big[h_E(\varpi,a') + 2h_E(\varpi,a) - h_E(\varpi,a) - h_E(\varpi,a') + h_E(\varpi,a)\big] + h_E(b,b'),\end{aligned}$$

and $h_E(b,b') = h_E(b, -b\sigma(a')/a) = h_E(b,\sigma a') - h_E(b,a) = v(b)\big[h_E(\varpi,a') - h_E(\varpi,a)\big]$. Altogether $\beta = v(b)\big[h_E(\varpi,a) + h_E(\varpi,a')\big]$, while $\kappa(X) + \kappa(X') = h_E(b,n) + h_E(b',n') = v(b)\big[2h_E(\varpi,a) + 2h_E(\varpi,a')\big]$. The sum $\kappa(X) + \kappa(X') + \beta = 3v(b)[\cdots] = 0 = \kappa(XX')$.

This exhausts all cases, since $x(\iota XX')$ is $\sigma(b)R$ when $b, b' \neq 0$ and the entries used in each case are the ones prescribed by (27). The last assertion is Corollary 10.3(d). □

Remark 10.5 (Checks and a second route)**.** (a) Script 12 evaluates (34) and (30) in the explicit model on 12,000 structured instances covering all five cases of the proof (including $v(1+\tau) \in \{1,\dots,5\}$ and $X' \in K^8,\dots,K^{12}$); all pass, and $\beta(X,X') \neq 0$ in about 44% of them, so the identity is not vacuous. Scripts 8 and 9 are earlier checks of the same identity through cube roots computed

by Hensel's lemma and on a stratified sample. None of these is part of the proof. (b) A second, independent argument reduces (H1) to a finite problem. The subgroup K^4 is a uniform pro-3 group of dimension 8, so central μ_3-extensions of K^4 are classified by alternating forms on K^4/K^8 [Laz, Ch. V; SW, §5]; the form of the cover is $\mathrm{Ad}(D^\times)$-invariant and vanishes on pairs of commuting elements by Part I, Proposition 3.1 and the depth bounds for the symbols of E_0 and of the ramified quadratic extensions of F; script 5a finds that the space of such forms is zero, so the cover splits over K^4 and the star section is a homomorphism on K^8. Script 5b then shows that every odd function on Q_6/Q_8 whose non-additivity is $D^\times$-equivariant is linear. Together these reduce (H1) to multiplicativity of ℓ on $K^6 \times K^8$, which is a special case of Proposition 10.4. We record this route because the Lazard-form computation is of independent interest for Target B.

11 The main theorem

Theorem 11.1 ([**P**^{c}] Target A)**.** *Let $F = \mathbb{Q}_3(\zeta_3)$, $n = 3$, D the quaternion division algebra over F, $\widetilde{H}$ its scalar-central 3-fold cover, and $\widetilde{\theta}$ a genuine central character compatible with the trivial character of $F^\times$. Then $L_D\mathbf{1} = \Theta_\tau$ for a unique genuine irreducible representation τ of $\widetilde{H}$, of dimension 27 and central character $\widetilde{\theta}$. Moreover:*

(a) $\tau|_{\widetilde{U}_1}$ is irreducible and equals $\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1} \lambda_0$, with λ_0 as in (25); $\tau|_{\widetilde{U}_6} = 27\,\psi_6$, and $\tau|_{\widetilde{U}_3}$ is the sum of the 27 distinct genuine characters of $\widetilde{U}_3/N_6$ whose restriction to $\widetilde{U}_4/N_6$ exhausts the genuine characters of that group.

(b) On the covered quadratic tori, τ restricts as in Proposition 6.2: $(3,2,2,2)$ on the unramified torus and $(3,2,1,1,1,1)$ on each ramified torus.

(c) Θ_τ is supported on the star lifts of good elements and of the center; its values are $27\zeta\widetilde{\theta}(\tilde{z})$ at $\zeta\tilde{z}t^*$ with $t \in F^\times K^2$, $9\zeta\widetilde{\theta}(\tilde{z})$ for $t \in F^\times K^1 \setminus F^\times K^2$, and $3\zeta\widetilde{\theta}(\tilde{z})$ for $t \notin F^\times K^1$.

(d) τ is the unique extension to $\widetilde{H}$ of $\tau|_{\widetilde{U}_1}$ with character values $3\zeta\widetilde{\theta}(\tilde{z})$ at the star lifts of the elements of 2-power order modulo the center.

Proof. (H1) holds by Proposition 10.4; Theorem 9.6 gives Target A together with (a), and Theorem 8.5 gives uniqueness and irreducibility of the restriction. The dependencies of this chain are itemized in Remark 11.2. (b) is Proposition 6.2; (c) collects (21) and the proof of Lemma 8.2; (d) follows from (c) since the four extensions of an $\widetilde{H}$-stable irreducible of $\widetilde{U}_1$ differ by the linear characters of D_8, which are distinguished by their values on P_0. □

Remark 11.2 (Dependencies of Theorem 11.1)**.** The proof uses the following inputs and nothing else.

(D1) [**E**] From Part I: the cover of $D^\times$ is the restriction along ι of the parameter-c_v Kazhdan–Patterson cover of $\mathrm{GL}_2(E_0)$ (Definition 2.2, Proposition 2.3); the commutator formula on covered tori (Proposition 3.1); good elements are $F^\times \cdot \{\text{cubes}\}$ in each torus (Proposition 3.5(a)); $E^\times/F^\times$ has no 3-torsion (Lemma 3.4); the star map (Theorem 3.9, Remark 8.1). From Part III: the normalization (1) of L_D with $b_D = 3$, $c_D = 27$, and the value $R = 1$ at elements of 2-power order.

(D2) [**E**] Kubota's cocycle on $\mathrm{GL}_2(E_0)$ (Lemma 10.1(a)), and the standard properties of the cubic Hilbert symbols of F and E_0 listed after Lemma 10.1.

(D3) [$\mathbf{P}^{\mathrm{c}}$] The explicit values of the two symbols (Appendix A), obtained as the unique Steinberg symbols by exact linear algebra (script 7). They enter the hand arguments only through $h_E(1+\varpi^3,\varpi)\neq 0$ (used to exclude $e=0$ in Proposition 9.5 via $\kappa_{1,5}(1,1)\neq 1$) and through the digit form of κ used by the scripts; the proof of Proposition 10.4 does not use them.

(D4) [$\mathbf{P}^{\mathrm{c}}$] Four exhaustive finite computations: the 72 elements of $\bar{G}_3$ from the 648 depth-one classes (script 1, also proved by hand in Proposition 7.2); the cube map on Q/Q_7 (script 2, corroborating Lemma 7.1); the actions on V and their orbits (script 3); and the design search over all odd D_8-invariant functions on the 36 lines together with the Heisenberg cocycle extension and the Fourier identities (scripts 4 and 6). Each is an enumeration of an explicitly described finite set.

(D5) [**P**] Everything else is proved by hand: the toral theorem and the three torus calculations, the positivity criterion, the depth lemma, the Brauer reduction (Theorem 8.5), the consequences of (H1) and the uniqueness of the design given the finite data (Section 9), the cube identity, and (H1) itself.

No sampled computation is load bearing. Scripts 5, 8–12 are corroboration (Remarks 10.5 and the next remark). Theorem 12.7 depends on no further computation; its Steps 2–3 and Proposition 12.8 are proved by hand from the items above.

Remark 11.3 (End-to-end verification)**.** Independently of the chain of reductions, the character of $\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}\lambda_0$ was computed in the explicit model of Section 10 from formula (25) (products of the eight D_8-conjugates, 27 coset representatives of $\widetilde{U}_1/\widetilde{U}_3$) and compared with (21) at star lifts of cubes of depth-one elements, at elements of $\widetilde{U}_6$ with all three values of ψ_6, at products $\sigma(g)\ell(k)\tilde{z}\zeta^m$, and at non-good elements of depths $3,4,5$; all values agree, including phases (scripts 11, 11b). Script 10 checks, again in the explicit model, that all 124 sampled coplanar triples of good lines have trivial defect and that all 186 sampled four-term relations $\sum v_i=0$ in $\bar{G}_3$ have defect $\sum e_*(v_i)$; that $\omega=0$ on 300 pairs; and that $\kappa_{1,5}(a,b)=\zeta^{-\mathrm{Tr}(a\bar{b})}$ on 300 pairs.

Corollary 11.4 ([$\mathbf{P}^{\mathrm{c}}$] Level-zero rows and their norm twists)**.** *Let ρ be one of the five irreducible representations of $D^\times$ trivial on K^1 (inflated from $D^\times/F^\times K^1\simeq D_8$), and let χ be a character of $F^\times$ trivial on μ_3. Then $L_D\Theta_{\rho\otimes(\chi\circ\mathrm{Nrd})}$, computed with any compatible genuine central character, is the character of a genuine irreducible representation of $\widetilde{H}$, of dimension $27\dim\rho$.*

Proof. First let $\chi=1$. On $\widetilde{U}_1$, $\Theta_\rho\circ p=\dim\rho$, so $L_D\Theta_\rho|_{\widetilde{U}_1}=(\dim\rho)\Phi$ is a character by Theorem 11.1; on Γ_{P_0} it is $3\,\psi_P\otimes\rho+3\dim\rho\sum_\pi(\dim\pi)\Theta_\pi$; and its restrictions to covered tori are characters by Lemma 5.1. The proof of Theorem 8.5 applies verbatim (Lemma 8.4 holds for every row by Part III), giving an irreducible character. For general χ, write $\chi=\chi_1^3$: this is possible because χ is trivial on μ_3, so that its restriction to $U_F^{(1)}\simeq\mu_3\times\mathbb{Z}_3^2$ factors through $\mathbb{Z}_3^2$, whose character group is 3-divisible, while on $\varpi^{\mathbb{Z}}\times\mu_2$ cube roots exist trivially. Twisting the representation τ_ρ just obtained by $\chi_1\circ\mathrm{Nrd}\circ p$ multiplies its character at $\tilde{z}t^*$ by $\chi_1(z)^2\chi(\mathrm{Nrd}\,t)$ and changes its central character to $\widetilde{\theta}':=\widetilde{\theta}\cdot(\chi_1^2\circ p)$, which is compatible with $\theta_\rho\chi^2=\theta_{\rho\otimes\chi\circ\mathrm{Nrd}}$ because $\widetilde{\theta}'(a^*)=\theta_\rho(a)\chi_1(a^3)^2$. The result is $L_D\Theta_{\rho\otimes\chi\circ\mathrm{Nrd}}$ computed with $\widetilde{\theta}'$; by Remark 8.6 any other compatible choice is obtained by a further twist. □

12 Target B: other rows

Throughout this section ρ is an irreducible smooth representation of $D^\times$ (finite dimensional, as $D^\times/F^\times$ is compact) with central character θ_ρ trivial on μ_3, θ_0 is a character of $F^\times$ with $\theta_0^3=\theta_\rho$,

and $\widetilde{\theta}(\zeta,z)=\zeta\theta_0(z)$ is the corresponding compatible genuine central character, in the coordinates of Lemma 10.1(b). We write $\phi_\rho := L_D\Theta_\rho$ and $\Phi_\rho := \phi_\rho|_{\widetilde{U}_1}$. For $\rho=\mathbf{1}$ we take, once and for all, $\theta_1 := \mathbf{1}$, i.e. $\widetilde{\theta}_1(\zeta,z)=\zeta$ (a compatible choice, by Remark 8.6; it is the normalization of the scripts), and write ϕ_1, $\widetilde{\theta}_1$, λ_0 for the objects of Sections 9–11 in this normalization. Two reductions are free.

Twisting. For a character χ of $F^\times$ trivial on μ_3, $\rho\otimes(\chi\circ\mathrm{Nrd})$ is irreducible with central character $\theta_\rho\chi^2$, and by the argument of Corollary 11.4, $L_D\Theta_{\rho\otimes\chi\circ\mathrm{Nrd}}$ is an irreducible character if and only if $L_D\Theta_\rho$ is. Squaring is surjective onto the characters of $F^\times=\varpi^{\mathbb{Z}}\times\mu_2\times U_F^{(1)}$ that are trivial at -1, so we may and do assume

$$\theta_\rho\in\{\mathbf{1},\ \mathrm{sgn}\},\qquad \mathrm{sgn}(\varpi^k\varepsilon u):=\varepsilon\quad(\varepsilon=\pm1,\ u\in U_F^{(1)}),\tag{35}$$

according to whether $\rho(-1)=+1$ or -1, and accordingly $\theta_0=\theta_1=\mathbf{1}$ or $\theta_0=\mathrm{sgn}$.

Unitarity. By Part III, $\langle\phi_\rho,\phi_\rho\rangle_{\widetilde{H}}=1$ and $\phi_\rho(1)=27\dim\rho>0$, so ϕ_ρ is an irreducible character as soon as it is a virtual character.

12.1 The reduction theorem for every row

Theorem 12.1 ([**P**] Reduction, all rows)**.** *$L_D\Theta_\rho$ is the character of a genuine irreducible representation of $\widetilde{H}$ if and only if Φ_ρ is a virtual character of $\widetilde{U}_1$.*

Proof. Only "if" needs proof. Assume (35). The finite quotient $\Gamma=\widetilde{H}/N$ of Section 8 is built from the finite row of ϕ_ρ exactly as before, with $N\cap\widetilde{Z}=\ker\widetilde{\theta}\cap\ker\widetilde{\theta}_1$ (so that both ϕ_ρ and ϕ_1 are functions on Γ); then $T=\widetilde{Z}N/N$ is μ_3 if $\theta_\rho=\mathbf{1}$ and $\mu_6=\mu_3\times\mu_2$ if $\theta_\rho=\mathrm{sgn}$; $\Gamma_0=\widetilde{U}_1/N$ is the Sylow 3-subgroup in the first case and is $\Gamma_0^{(3)}\times\mu_2$ with $\Gamma_0^{(3)}$ the Sylow 3-subgroup in the second; in both cases Γ/Γ_0 is D_8. We check the three inputs of the proof of Theorem 8.5.

Tori. $\phi_\rho|_{\widetilde{E^\times}}$ is a genuine representation character for every maximal subfield E (Lemma 5.1: $\Theta_\rho|_{E^\times}$ is a sum of characters α with $\alpha|_{F^\times}=\theta_\rho$), so Lemma 8.1 holds verbatim.

Centralizers. Lemma 8.3 concerns Γ_0 and the images of tori and is unchanged; the μ_2 in T is central.

Two-groups. Let $\hat{P}_0$ be the preimage of $P_0F^\times$ and $\Gamma_{P_0}=\hat{P}_0N/N$, a central extension of D_8 by T; it contains a Sylow 2-subgroup of Γ. Every $\eta\in\hat{P}_0\subset D^\times$ is good: $\eta^4\in F^\times$, so $\eta=\eta^4\cdot(\eta^{-1})^3$. Put $x_\eta:=\widetilde{\eta^4}\,(\tilde{\eta}^{-1})^3$, a lift of η with $x_{\eta c}=x_\eta\tilde{c}$ for $c\in F^\times$. By (1) and $R=1$ off $F^\times K^1$,

$$\phi_\rho(x_\eta)=3\,\theta_0(\eta^4)\,\Theta_\rho(\eta^{-1})=3\,\theta_0(\eta^4)\,\Theta_{\rho^\vee}(\eta)\qquad(\eta\notin F^\times),$$

and $\phi_\rho=27\dim\rho\cdot\widetilde{\theta}$ on $\widetilde{Z}$. Let ψ_P be the genuine character of order 3 of Γ_{P_0} trivial on a Sylow 2-subgroup (unique, by Schur–Zassenhaus for the μ_3-part of T); on the image of $\widetilde{Z}$ it is $\widetilde{\theta}_1$, i.e. $\psi_P(z,0)=\theta_1(z)$, and therefore $\psi_P(x_\eta)=\theta_1(\eta^4)\psi_P(\tilde{\eta}^{-1})^3=\theta_1(\eta^4)$. (We do not twist $\rho^\vee$ by $\theta_1^2\circ\mathrm{Nrd}$: on scalars that twist is $\theta_1(z^2)^2=\theta_1(z)$, not trivial.)

If $\theta_\rho=\mathbf{1}$, then $\theta_0=\theta_1$, so $\phi_\rho(x_\eta)=3\psi_P(x_\eta)\Theta_{\rho^\vee}(\eta)$, and $\rho^\vee$ is trivial on $F^\times$, so $\Theta_{\rho^\vee}$ is the character of a representation of $D_8=\hat{P}_0/F^\times$. With $\Gamma_{P_0}=\mu_3\times\tilde{P}_0$ as in Lemma 8.2,

$$\phi_\rho|_{\Gamma_{P_0}}=\psi_P\otimes\Big(3\,\rho^\vee|_{D_8}+3\dim\rho\sum_{\pi\in\mathrm{Irr}(D_8)}(\dim\pi)\,\Theta_\pi\Big),$$

a character (the second term is $24\dim\rho$ times the delta function at $1\in D_8$).

If $\theta_\rho = \mathrm{sgn}$, then $\theta_0 = \mathrm{sgn}\cdot\theta_1$ and $\phi_\rho(x_\eta) = 3\psi_P(x_\eta)\,\mathrm{sgn}(\eta^4)\,\Theta_{\rho^\vee}(\eta)$, where now $\rho^\vee$ has central character sgn. Here $\Gamma_{P_0} = \mu_3 \times \Gamma''$ with $\Gamma'' := \Gamma_{P_0}/\mu_3$, the quotient of $\hat{P}_0$ by μ_3 and the lifts of $\ker\mathrm{sgn} \subset F^\times$, a central extension of D_8 by μ_2; $\rho^\vee|_{\hat{P}_0}$ factors through Γ''. With a, b the images of $1+i$ and j one has $a^8 = 1$ (as $(1+i)^4 = -4$ and $\mathrm{sgn}(-4) = -1$), $b^2 = 1$ and $bab^{-1} = a^3$ (as $j(1+i)j^{-1} = 2(1+i)^{-1}$ and $\mathrm{sgn}(2) = -1$), so $\Gamma'' \simeq SD_{16}$, the semidihedral group, with a^4 the image of -1. The function $f(\eta) := \mathrm{sgn}(\eta^4)$ is -1 on the classes $\{a, a^3\}, \{a^5, a^7\}$ and $+1$ elsewhere. Having central character sgn, $\rho^\vee|_{\Gamma''}$ is a sum of copies of the two faithful two-dimensional characters χ_1, χ_2 of SD_{16} ($a \mapsto \pm i\sqrt{2}$, $a^2 \mapsto 0$, $a^4 \mapsto -2$, 0 off $\langle a\rangle$); a direct check on the seven classes gives $f\chi_1 = \chi_2$ and $f\chi_2 = \chi_1$. Hence $f\cdot\Theta_{\rho^\vee}$ is a character of Γ'', and with δ^{sgn} the function ± 1 at a^0, a^4 and 0 elsewhere, $8\delta^{\mathrm{sgn}} = 2\chi_1 + 2\chi_2$, so

$$\phi_\rho|_{\Gamma_{P_0}} = \psi_P \otimes \left(3f\,\Theta_{\rho^\vee|_{\Gamma''}} + 3\dim\rho\,(2\chi_1 + 2\chi_2)\right)$$

is a character.

Brauer's theorem. Let $B = B_2 \times B_3 \subset \Gamma$ be elementary, and let $\bar{B}_2$ be the image of B_2 in $\Gamma/\Gamma_0 = D_8$. (When $T \supset \mu_2$, B_2 may be noncyclic while $\bar{B}_2$ is cyclic, e.g. $B_2 = \langle a^4, b\rangle$; the case distinction must therefore be made on $\bar{B}_2$.) If $\bar{B}_2 = 1$, then $B \subset \Gamma_0$ and $\phi_\rho|_B = \Phi_\rho|_B$ is a virtual character by hypothesis. If $\bar{B}_2 = \langle\bar{y}\rangle$ is cyclic and nontrivial, pick $y \in B_2$ over $\bar{y}$; then $B_2 \subset \langle y\rangle\cdot(B_2 \cap T)$, $B_3 \subset C_{\Gamma_0}(y) \subset T_{E_y}$ by Lemma 8.3(a), $y \in T_{E_y}$ and $T \subset T_{E_y}$, so $B \subset T_{E_y}$ and Lemma 8.1 applies. If $\bar{B}_2$ is noncyclic, then $B_3 \subset C_{\Gamma_0}(B_2) \subset T$ by Lemma 8.3(b) (whose proof uses only the images of the elements of B_2 in D_8), so $B \subset B_2T$, which is conjugate into Γ_{P_0} because Γ_{P_0} contains a Sylow 2-subgroup; the two-group computation applies.

Brauer's theorem now gives that ϕ_ρ is a virtual character of Γ, and unitarity finishes the proof. □

12.2 The cube-root transform

Lemma 12.2 ([**P**] Product formula)**.** *Assume* (35)*. There is a unique function F_ρ on the set of star lifts of good elements of H such that*

$$\phi_\rho = \phi_1\cdot F_\rho \quad \text{on that set,} \qquad F_\rho(\zeta\tilde{z}\,t^*) = \theta_0\theta_1^{-1}(z)\,\Theta_\rho(t), \tag{36}$$

and both sides of (36) *vanish elsewhere. F_ρ is invariant under conjugation by $\widetilde{H}$ and under multiplication by μ_3. On the star lifts of good elements of U_3,*

$$F_\rho(\tilde{z}\,t^*) = \varepsilon_\rho(z)\,\Theta_\rho(t) \qquad (zt^3 \in U_3,\ t \in K^1), \tag{37}$$

where $\varepsilon_\rho = \mathbf{1}$ if $\theta_\rho = \mathbf{1}$ and $\varepsilon_\rho = \mathrm{sgn}$ if $\theta_\rho = \mathrm{sgn}$; here $t \in K^1$ is determined by zt^3 up to $U_F^{(1)}$, on which Θ_ρ is invariant, and z up to $(U_F^{(1)})^3 \subset \ker\varepsilon_\rho$. In particular $g \mapsto \varepsilon_\rho(z)$ is the character of $U_3 = F^\times K^3$ trivial on K^3 and equal to ε_ρ on $F^\times$ (well defined because $F^\times \cap K^3 = U_F^{(2)}$), and it is $\mathrm{Ad}(U_1)$-invariant.

Proof. Both ϕ_ρ and ϕ_1 vanish off star lifts of good elements, and there $\phi_\rho(\zeta\tilde{z}t^*) = 3\zeta\theta_0(z)R(t)\Theta_\rho(t)$, $\phi_1(\zeta\tilde{z}t^*) = 3\zeta\theta_1(z)R(t)$ with $R(t) \neq 0$. If $\tilde{z}t^* = \tilde{z}'t'^*$ then $t' = ct$ with $c \in F^\times$ and $z' = zc^{-3}$ (as in Lemma 7.3), and $\theta_0\theta_1^{-1}(c^{-3})\Theta_\rho(ct) = \theta_\rho(c)^{-1}\theta_\rho(c)\Theta_\rho(t)$, so F_ρ is well defined. Invariance is clear. For $zt^3 \in U_3$ with $t \in K^1$: $\theta_0\theta_1^{-1} = \varepsilon_\rho$ on $F^\times$ by the normalization, and $c \in F^\times \cap K^1 = U_F^{(1)}$, on which $\theta_\rho = 1$. □

Remark 12.3 (Adams operations)**.** F_ρ restricted to $\widetilde{U}_1$ is the Adams-type transform $\psi^{1/3}\Theta_\rho$, "Θ_ρ at the cube root," corrected by ε_ρ. At a tame place $\psi^{1/3}$ is a Galois conjugation of characters

(Part III, Lemma 7.5), which is why Target B is trivial there; at the wild place it is not even a virtual character in general, and the content of the next criterion is that it need not be: it must only be compatible with the character λ_0 on the good set.

Proposition 12.4 ([**P**] A sufficient criterion)**.** *Assume* (35)*. Suppose there is a virtual character Ψ of U_3 (a $\mathbb{Z}$-combination of characters of finite-dimensional smooth representations) which is $\mathrm{Ad}(U_1)$-invariant and satisfies $\Psi(zt^3) = F_\rho(\tilde{zt}^*)$ for all good $zt^3 \in U_3$. Then $L_D\Theta_\rho$ is the character of a genuine irreducible representation τ_ρ of $\widetilde{H}$ of dimension $27\dim\rho$, and*

$$\Theta_{\tau_\rho}|_{\widetilde{U}_1} = \sum_i n_i\,\Theta_{\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}(\lambda_0\otimes\sigma_i\circ p)}, \qquad \Psi = \sum_i n_i\Theta_{\sigma_i},\ n_i\in\mathbb{Z}. \tag{38}$$

Proof. For a finite-dimensional representation σ of U_3 with $\mathrm{Ad}(U_1)$-invariant character, the induced character satisfies, for $x \in \widetilde{U}_3$ and g running over $\widetilde{U}_1/\widetilde{U}_3$,

$$\Theta_{\mathrm{Ind}(\lambda_0\otimes\sigma\circ p)}(x) = \sum_g \lambda_0(gxg^{-1})\Theta_\sigma(p(gxg^{-1})) = \Theta_\sigma(p(x))\sum_g \lambda_0(gxg^{-1}) = \Theta_\sigma(p(x))\,\Theta_{\mathrm{Ind}\,\lambda_0}(x),$$

and both sides vanish off $\widetilde{U}_3$. By Theorem 11.1(a), $\Theta_{\mathrm{Ind}\,\lambda_0} = \Phi_1$, which is supported on star lifts of good elements of U_3 (Lemma 7.3). Summing with the n_i and using (36), $\sum_i n_i\Theta_{\mathrm{Ind}(\lambda_0\otimes\sigma_i\circ p)} = \Phi_1\cdot(\Psi\circ p) = \Phi_1 F_\rho = \Phi_\rho$ on $\widetilde{U}_1$. So Φ_ρ is a virtual character of $\widetilde{U}_1$, and Theorem 12.1 applies. □

12.3 All representations trivial on K^2

Let $\pi_3 : U_3 \to U_3/U_4 \simeq Q_3/Q_4 \simeq \mathbb{F}_9$ be the depth-three digit (the A-coordinate of (19)), a homomorphism which is $\mathrm{Ad}(U_1)$-invariant because $[U_1, U_3] \subset U_4$, and identify $K^1/K^2 \simeq \mathbb{F}_9$ by $1 + wj \mapsto w$.

Theorem 12.5 ([**P**] Target B for level at most one)**.** *Let ρ be an irreducible representation of $D^\times$ trivial on K^2, with $\theta_\rho|_{\mu_3} = 1$ (automatic, since $\mu_3 \subset U_F^{(1)} \subset K^2$). Then $L_D\Theta_\rho$, computed with any compatible genuine central character, is the character of a genuine irreducible representation τ_ρ of $\widetilde{H}$ of dimension $27\dim\rho$. After the normalization* (35)*,*

$$\tau_\rho|_{\widetilde{U}_1} = \bigoplus_\psi m_\psi\ \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}(\lambda_0\otimes(\varepsilon_\rho\cdot\psi\circ\pi_3)\circ p), \qquad \rho|_{K^1} = \bigoplus_\psi m_\psi\,\psi,$$

the sum over the characters ψ of $K^1/K^2 \simeq \mathbb{F}_9$; the induced representations occurring are irreducible and pairwise distinct. The same conclusion holds for $\rho\otimes(\chi\circ\mathrm{Nrd})$ for every character χ of $F^\times$ trivial on μ_3; these are all the irreducible representations of $D^\times$ whose restriction to K^2 is a multiple of a character of the form $\chi\circ\mathrm{Nrd}$.

Proof. By twisting we may assume (35). Let $g = zt^3$ be a good element of U_3 with $t \in K^1$. If $t \in F^\times K^2$, then $g \in U_6$, $\pi_3(g) = 0$ and $\Theta_\rho(t) = \dim\rho = \Theta_{\rho|_{K^1}}(0)$. Otherwise $t \equiv 1 + wj$ modulo K^2 with $w \neq 0$, and by Lemma 7.1(c) the depth-three digit of g is w^5; since $w^4 = \mathrm{N}_{\mathbb{F}_9/\mathbb{F}_3}(w) = \pm 1$, $w^5 = \pm w$, so $w = \pm\pi_3(g)$. Because Θ_ρ is a class function and $i(1+wj)i^{-1} = 1 - wj$, we get $\Theta_\rho(t) = \Theta_{\rho|_{K^1}}(\pi_3(g))$ in both cases. By (37), $F_\rho(\tilde{zt}^*) = \varepsilon_\rho(z)\Theta_{\rho|_{K^1}}(\pi_3(g))$, which is the value at g of the character of the representation $\Psi := \varepsilon_\rho\otimes(\rho|_{K^1})\circ\pi_3$ of U_3 (with $\rho|_{K^1}$ regarded as a representation of K^1/K^2). Its character is $\mathrm{Ad}(U_1)$-invariant, so Proposition 12.4 applies with all $n_i = m_\psi \geq 0$, giving the displayed decomposition. The restriction $\tau_\rho|_{\widetilde{U}_1}$ has norm $\int_{U_1/Z}|\Theta_\rho|^2\,d\mu$ by the Weyl-integration computation of Lemma 8.4 (substituting $t = s^3$, $|\phi_\rho|^2(s^3) = 9R(s)^2|\Theta_\rho(s)|^2$), and this

is $\langle \Theta_\rho|_{K^1}, \Theta_\rho|_{K^1}\rangle = \sum_\psi m_\psi^2$. On the other hand $\langle \sum m_\psi\chi_\psi, \sum m_\psi\chi_\psi\rangle \geq \sum m_\psi^2\langle\chi_\psi,\chi_\psi\rangle \geq \sum m_\psi^2$ for the characters χ_ψ of the induced representations, with equality only if each χ_ψ with $m_\psi > 0$ is irreducible and they are pairwise orthogonal. The last sentence: if $\rho|_{K^2}$ is a multiple of $\chi\circ\mathrm{Nrd}$, then $\rho\otimes(\chi^{-1}\circ\mathrm{Nrd})$ is trivial on K^2; and χ may be modified by any character trivial on $\mathrm{Nrd}(K^2) = U_F^{(1)}$ without changing this. □

Remark 12.6. (a) The theorem contains Corollary 11.4 and extends it to the level-zero representations with $\rho(-1) = -1$ (the two-dimensional representations of $D^\times/K^1$ induced from the characters of μ_8 of odd order), which are not twists of representations of D_8. (b) For the four four-dimensional irreducible representations of $\mathbb{F}_9 \rtimes D_8$ the theorem gives $\dim\tau_\rho = 108$ and $\tau_\rho|_{\widetilde{U}_1}$ a sum of four irreducibles of dimension 27, permuted transitively by D_8. (c) Nothing in the proof used irreducibility of ρ beyond the last sentence: for every finite-dimensional representation ρ of $D^\times$ trivial on K^2, $L_D\Theta_\rho$ is the character of a genuine representation of $\widetilde{H}$.

12.4 Level two

Let now ρ be irreducible and trivial on K^3 but not on K^2. Normalize $\theta_\rho \in \{\mathbf{1}, \mathrm{sgn}\}$ as in (35); then $\rho|_{K^1}$ factors through $\mathrm{Heis} := Q_1/Q_3 = K^1/K^3U_F^{(1)}$, the Heisenberg group of order 27, and is nontrivial on its center Q_2/Q_3. The group $D_8 = D^\times/F^\times K^1$ permutes the two three-dimensional irreducible characters χ_3, χ_3' of Heis, so $\rho|_{K^1} = e(\chi_3 + \chi_3')$, and $e = 1$ for irreducible ρ: the stabilizer of χ_3 in D_8 is the cyclic group of order 4 generated by the class of $1+i$, which acts trivially on the center Q_2/Q_3 of Heis (while j inverts it), and a character stable under a cyclic group extends. So $\dim\rho = 6$; there are four such ρ for each normalized central character. For these ρ the criterion of Proposition 12.4 *fails*: the restriction of F_ρ to the good classes of Q_3/Q_7 is a rational but not an integral combination of the restrictions of the $\mathrm{Ad}(Q_1)$-orbit sums of $\mathrm{Irr}(Q_3/Q_7)$ (script 13). Nevertheless:

Theorem 12.7 ([$\mathbf{P}^c$] Target B for level two)**.** *Let ρ be an irreducible representation of $D^\times$ trivial on K^3 with $\theta_\rho|_{\mu_3} = 1$. Then $L_D\Theta_\rho$, computed with any compatible genuine central character, is the character of a genuine irreducible representation τ_ρ of $\widetilde{H}$ of dimension $27\dim\rho$. If $\rho|_{K^2}$ is not scalar, then after normalizing $\theta_\rho \in \{\mathbf{1}, \mathrm{sgn}\}$,*

$$\tau_\rho|_{\widetilde{U}_1} = (\varepsilon_\rho\circ p)\otimes\left(\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}(\lambda_0\otimes\chi_1'\circ p) \;\oplus\; \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1}(\lambda_0\otimes\chi_2'\circ p)\right),$$

where χ_1', χ_2' are three-dimensional irreducible characters of Q_3/Q_7 (one from each of the two orbits described in the proof); both induced representations are irreducible of dimension 81, and they are distinct.

Proof. By Theorem 12.5 we may assume $\rho|_{K^2}$ nonscalar, and by twisting, (35). By Theorem 12.1 it suffices to show that $\Phi_\rho = \Phi_1 F_\rho$ is a virtual character of $\widetilde{U}_1$. The proof has three steps, the last two being exact finite computations.

Step 1: a finite quotient. For $g = zt^3$ good in U_3 with $t \in K^1$, the class of t in Heis depends only on the class of g in Q_3/Q_7: for $g \in Q_6$ this is the linearity of the inverse Artin–Schreier map (Lemma 7.1(d)), and for $g \in G_3$ it follows from the proof of Lemma 7.1(f) ($(t(1+u))^3 \in t^3Q_6$ for $u \in \mathfrak{P}^3$); script 13 verifies it on the 219 good classes of Q_3/Q_7 (216 of depth three, 3 in Q_6/Q_7), and verifies that the cube root is $\mathrm{Ad}(Q_1)$-equivariant. Hence $F_\rho(\tilde{z}t^*) = \varepsilon_\rho(z)\Theta_\rho(t)$ (Lemma 12.2) is a function of the class of g in Q_3/Q_7 (times the character ε_ρ of U_1). Moreover Φ_ρ is invariant under right multiplication by $\ell(K^7)$: for $n \in K^7$ and g good in U_3, $\ell(gn) = \ell(g)\ell(n)$ (for $g \in U_6$ by (H1); for $g \in G_3$ by Lemma 9.3(d), since $\ell(n) \in N_6$ as $\theta_0(\mathrm{Nrd}\,n) = 1$), and gn is good (Lemma 7.1(f)).

So Φ_ρ is a class function on the finite group $\hat\Gamma := \widetilde{U}_1/\ell(K^7)\ker\widetilde\theta$, of order 3^{10}, supported on the normal subgroup $\hat H := \widetilde{U}_3/\ell(K^7)\ker\widetilde\theta$ of index 27, which is a central extension of $H := Q_3/Q_7$ (order 3^6) by μ_3. (In the normalization $\theta_1 = \mathbf{1}$ fixed above, $\ker\widetilde\theta = \ker\widetilde\theta_1 = \{(1,z)\}$ when $\theta_\rho = \mathbf{1}$; when $\theta_\rho = \mathrm{sgn}$ we use $\ker\widetilde\theta_1$ in place of $\ker\widetilde\theta$ and carry the character $\varepsilon_\rho\circ p$ of $\widetilde{U}_1$ separately, as in Remark (b) below.)

Step 2: the restriction to $\widetilde{U}_3$. By Theorem 11.1(a), $\Phi_1|_{\widetilde{U}_3} = \sum_{t\in\widetilde{U}_1/\widetilde{U}_3}\lambda_0^t = \lambda_0\cdot(D\circ p)$ with $D(v) = \sum_{h\in\Lambda}\zeta^{h(v)}$, a function on $V = Q_3/Q_6$ which is 27 at $v=0$, $9\zeta^{-e_*(v)}$ on $\bar G_3$ and 0 elsewhere (Proposition 9.4). Hence

$$\Phi_\rho|_{\widetilde{U}_3} = (\varepsilon_\rho\circ p)\,\lambda_0\cdot\big((\Theta_\rho\circ r)\,D\big)\circ p,$$

where $(\Theta_\rho\circ r)D$ is a class function on H supported on the good classes. We compute it by hand. The structure of H: its derived subgroup is $H' = Q_6/Q_7 \simeq \mathbb{F}_9/\mathbb{F}_3$, of order 3 ($[K^3,K^3]\subset K^6$, and the bracket of leading terms $[\varpi aj,\varpi bj] = \varpi^3(a\bar b - b\bar a)$ is nonzero modulo $\mathbb{F}_3$ for suitable a,b); its center is Q_4/Q_7 ($[K^3,K^4]\subset K^7$, and the same bracket shows that no element of depth exactly 3 is central), of index 9. So H has $|H/H'| = 243$ linear characters, all its nonlinear irreducible characters have degree $\le\sqrt{[H:Z(H)]} = 3$, hence exactly 3, and there are $(729-243)/9 = 54$ of them; they are the irreducible characters nontrivial on H', and each restricts to H' as 3ω for one of the two nontrivial characters ω of H', 27 for each ω. The isotypic decomposition of the regular character gives, for $h\in H$,

$$\sum_{\chi'(1)=3,\ \chi'|_{H'}=3\omega}\chi'(h) = 81\,\omega(h)\,\mathbf{1}_{H'}(h), \qquad \sum_{\chi'(1)=3}\chi'(h) = \begin{cases}162, & h=1,\\ -81, & h\in H'\setminus 1,\\ 0, & h\notin H'.\end{cases} \tag{39}$$

On the other side, let χ_3 be one of the two three-dimensional characters of Heis, with central character ω_3 on Q_2/Q_3; it vanishes off the center. For a good class g of depth exactly three, $r(g)\in\mathrm{Heis}$ has depth one, so $(\chi_3\circ r)(g) = 0$. For $g\in Q_6/Q_7 = H'$ one has $D(g) = 27$, and r restricted to H' is the inverse of the Artin–Schreier bijection $Q_2/Q_3\to Q_6/Q_7$ of Lemma 7.1(d), an isomorphism of groups of order 3; so $(\chi_3\circ r)(g) = 3\omega_3(r(g))$ and $(\chi_3\circ r)D = 81\,(\omega_3\circ r)\mathbf{1}_{H'}$. Comparing with (39),

$$(\chi_3\circ r)\cdot D = \sum_{\chi'(1)=3,\ \chi'|_{H'}=3\,\omega_3\circ r}\chi', \qquad ((\chi_3+\chi_3')\circ r)\cdot D = \sum_{\chi'(1)=3}\chi', \tag{40}$$

each of the 54 characters occurring exactly once. (Script 14 confirms this from the character table of H; it also returns, for the orbit sums of linear characters of Heis, the sums of 27, resp. 108, linear characters of H, in accordance with Theorem 12.5.) Thus $\Phi_\rho|_{\widetilde{U}_3} = (\varepsilon_\rho\circ p)\sum_{\mu\in S}\mu$, where S is the set of the 54 genuine irreducible characters $\mu = \lambda_0\cdot\chi'\circ p$ of $\hat H$ of degree 3 with central character ζ on μ_3: a multiplicity-free character of dimension 162.

Step 3: Clifford theory. A class function on $\hat\Gamma$ supported on the normal subgroup $\hat H$ is determined by its restriction to $\hat H$, and for $\mu\in\mathrm{Irr}(\hat H)$ with $\hat\Gamma$-orbit O and stabilizer $\hat\Gamma_\mu$, the character of $\mathrm{Ind}_{\hat H}^{\hat\Gamma}\mu$ is supported on $\hat H$ with restriction $[\hat\Gamma_\mu:\hat H]\sum_{\mu'\in O}\mu'$. By Proposition 12.8 below, $\mathrm{Ad}(Q_1)$ acts freely on the genuine irreducible characters of $\hat H$, so S is the union of two orbits O_1, O_2 of size 27 (one for each ω, by (40) and the invariance of the central character), and

$$\Phi_\rho = (\varepsilon_\rho\circ p)\big(\Theta_{\mathrm{Ind}\,\mu_1} + \Theta_{\mathrm{Ind}\,\mu_2}\big), \qquad \mu_i\in O_i,$$

with $\mathrm{Ind}_{\hat H}^{\hat\Gamma}\mu_i$ irreducible of dimension 81 and distinct. (Had some orbit $O\subset S$ had size $|O|<27$, then for an irreducible constituent χ of $\mathrm{Ind}\,\mu_O$ with $\chi|_{\hat H} = e\sum_O\mu'$ one would have $\langle\Phi_\rho,\chi\rangle = e|O|/27\notin\mathbb{Z}$,

since $e^2 \le [\hat\Gamma_\mu : \hat H] = 27/|O|$.) This proves the displayed decomposition, and $\dim \tau_\rho = 162 = 27 \cdot 6$. □

Proposition 12.8 ([**P**^{c}] Freeness)**.** $\mathrm{Ad}(\widetilde U_1)$ *acts freely on the set of genuine irreducible characters of* $\hat H = \widetilde U_3/\ell(K^7)\ker\widetilde\theta_1$*: for every genuine irreducible* μ *and every* $t \in U_1 \setminus U_3$, $\mu \circ \mathrm{Ad}(t) \ne \mu$. *Equivalently,* $\mathrm{Ind}_{\widetilde U_3}^{\widetilde U_1}\mu$ *is irreducible for every genuine irreducible representation* μ *of* $\widetilde U_3$ *trivial on* $\ell(K^7)\ker\widetilde\theta_1$.

Proof. Let $A := Q_4/Q_7$, central in H, and $\hat A$ its preimage in $\hat H$. $\hat A$ is central in $\hat H$: for $x \in K^3$, $y \in K^4$ one has $[x,y] \in K^7$, so $[\tilde x, \tilde y] = \zeta^a \ell([x,y])$ with $\zeta^a = \psi_6([\tilde x,\tilde y]) = \omega(x,y) = 1$ by Lemma 9.3(c), and $\ell([x,y]) \in \ell(K^7)$. Hence $\mu|_{\hat A} = \mu(1)\,\hat\nu$ for a genuine character $\hat\nu$ of $\hat A$, and $\mu \circ \mathrm{Ad}(t) = \mu$ implies $\hat\nu \circ \mathrm{Ad}(t) = \hat\nu$. Write $\hat\nu = \lambda_0|_{\hat A} \cdot (\alpha \circ p)$ with α a character of A, and let $\omega_\mu := \alpha|_{H'}$ (possibly trivial). Since $\lambda_0(\ell(u)) = 1$ for $u \in U_6$ ($\ell(u) \in N_6$, as $\psi_6(\ell(u)) = \theta_1(z_u) = 1$), the restriction of $\hat\nu$ to $\ell(Q_6/Q_7)$ is ω_μ. For $y \in A$,

$$\frac{\hat\nu \circ \mathrm{Ad}(t)}{\hat\nu}(\tilde y) = \hat\nu([\tilde t,\tilde y]) = \frac{\lambda_0^t}{\lambda_0}(\tilde y)\cdot\omega_\mu([t,y]) = c_t(y)\,\omega_\mu([t,y]),$$

where $c_t := \lambda_0^t/\lambda_0 = \zeta^{h_t}$ is a character of V with $h_t \in \Lambda$, and $t \mapsto h_t$ is a bijection $Q_1/Q_3 \to \Lambda$ with $h_t = 0$ iff $t \in Q_3$ (because $\sum_t \lambda_0^t = \Phi_1|_{\widetilde U_3} = \sum_{h\in\Lambda}\lambda_h$, Theorem 11.1(a), and the 27 characters λ_h are distinct). Let $t \notin Q_3$.

If $t \in Q_2$, then $[t,y] \in K^7$ for all $y \in K^4$ (the leading brackets $[\varpi a, \varpi^2 c] = 0$ and $[\varpi a, \varpi^2 dj] = \varpi^3 d(a-\bar a)j$ have depth ≥ 7), so the displayed ratio is $c_t|_A$, which is nontrivial: no nonzero element of Λ vanishes on the coordinates $(C_{\mathrm{im}}, B_{\mathrm{re}}, B_{\mathrm{im}})$ of Q_4/Q_6 (Appendix A).

If t has depth exactly one with leading coefficient $a \in \mathbb{F}_9^\times$, take $y \in Q_5$ with leading coefficient b (so $y = 1 + \varpi^2 bj + \cdots$). Then $c_t(y) = \lambda_0([\tilde t,\tilde y]) = \zeta^{\kappa_{1,5}(t,y)} = \zeta^{\beta\,\mathrm{Tr}(a\bar b)}$ with $\beta \ne 0$ (Lemma 9.3(b)), and $[t,y] \in Q_6/Q_7 \simeq \mathbb{F}_9/\mathbb{F}_3$ is the class of $\varpi^3(a\bar b - \bar a b) = 2i\,\mathrm{Im}(a\bar b)\varpi^3$, so $\omega_\mu([t,y]) = \zeta^{2\gamma\,\mathrm{Im}(a\bar b)}$ for some $\gamma \in \mathbb{F}_3$ depending on μ. As b runs over $\mathbb{F}_9$ so does $u = a\bar b$, and $u \mapsto 2\beta\,\mathrm{Re}(u) + 2\gamma\,\mathrm{Im}(u)$ is a nonzero $\mathbb{F}_3$-linear functional. So the ratio is nontrivial on Q_5.

In both cases $\hat\nu \circ \mathrm{Ad}(t) \ne \hat\nu$, hence $\mu \circ \mathrm{Ad}(t) \ne \mu$. □

Remark 12.9 (Certificate)**.** Script 15 computes λ_0 on $\hat H$ in the cocycle model, the twisted action $\mu^t = \lambda_0 c_t(\chi' \circ \mathrm{Ad}\,t)$ of the 27 coset representatives on the 54 characters of S and on the 243 genuine linear characters, verifies that no nontrivial t fixes any of them, and writes the orbit table to `level2_orbit_certificate.txt`: two orbits of size 27 on S, nine of size 27 on the linear characters (the nine induced representations of Theorem 12.5). For comparison, the untwisted action $\chi' \mapsto \chi' \circ \mathrm{Ad}\,t$ on the 54 characters of H has six orbits of size 9: the λ_0-twist is what makes the action free. These computations corroborate Proposition 12.8 and are not used in its proof.

Remark 12.10. (a) The criterion of Proposition 12.4 is sufficient but not necessary: here $\tau_\rho|_{\widetilde U_1}$ *is* of the form $\mathrm{Ind}(\lambda_0 \otimes \sigma)$ with $\sigma = \chi_1' + \chi_2'$, but Θ_σ is not $\mathrm{Ad}(U_1)$-invariant, and the product formula $\Theta_{\mathrm{Ind}(\lambda_0\otimes\sigma)} = \Theta_\sigma\Theta_{\mathrm{Ind}\,\lambda_0}$ fails. (b) The ε_ρ bookkeeping: for $\theta_\rho = \mathrm{sgn}$, ε_ρ is a character of U_1 trivial on K^1, so twisting by $\varepsilon_\rho \circ p$ preserves virtual characters of $\widetilde U_1$ and commutes with everything above. (c) Label. The theorem rests on Target A through λ_0, Λ and e_* only (Theorem 11.1(a), Proposition 9.5, and the table of Λ); Steps 2 and 3 are hand arguments. Scripts 13–15 are corroboration.

12.5 Level at least three

[**O**] We record what the two level theorems suggest, and what they do not. For $m \ge 2$ put $\mathrm{Heis}_m := Q_1/Q_{m+1}$, $H_m := Q_3/Q_{3m+1}$, $\hat H_m := \widetilde U_3/\ell(K^{3m+1})\ker\widetilde\theta_1$ and $\hat\Gamma_m := \widetilde U_1/\ell(K^{3m+1})\ker\widetilde\theta_1$

(for $m=1$ use K^6 in place of K^4, since ℓ is defined on K^6; then $\hat{H}_1 = \widetilde{U}_3/N_6 = \widetilde{V}$ and the genuine characters of $\hat{H}_1$ are the 243 characters λ_h, $h \in V^\vee$). The cube root induces a well-defined *surjection* r from the good classes of H_m onto Heis_m, many-to-one in general (219 good classes of H_2 onto the 27 elements of Heis_2); the fibers are the sets $\{t^3 Q_{3m+1}\}$ for t in a fixed class of Q_1/Q_{m+1}.

The proofs of Theorems 12.5 and 12.7 both run as follows: $\Phi_\rho|_{\widetilde{U}_3} = \lambda_0 \cdot ((\Theta_\rho \circ r)D)$, and (i) $(\Theta_\rho \circ r)D$ is a character of H_m, (ii) $\mathrm{Ad}(Q_1)$ acts freely on the genuine irreducible characters of $\hat{H}_m$, whence Φ_ρ is a sum of irreducible induced characters. Statement (ii) concerns the cover alone; it is Proposition 12.8 for $m=2$ and is immediate for $m=1$ (the λ_h form a single free Q_1/Q_3-orbit, Proposition 9.5). Statement (i) was proved for $m \le 2$ by exhibiting $(\Theta_\rho \circ r)D$ as a sum of irreducible characters of H_m: for linear χ it is $\sum_{h\in\Lambda} \zeta^h \cdot (\chi \circ \pi_3)$, for χ_3 it is (40). Together they give, for every ρ trivial on K^{m+1} with $\rho|_{K^1} = \sum_\chi m_\chi \chi$,

$$\tau_\rho|_{\widetilde{U}_1} = (\varepsilon_\rho \circ p) \otimes \bigoplus_i \mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1} \mu_i,$$

the μ_i running over one representative of each Q_1-orbit of genuine constituents of $\lambda_0(\Theta_\rho \circ r)D$.

Two warnings about the shape of a general statement. First, the correspondence is not constituent-wise for linear χ: for a single additive character χ of $\mathbb{F}_9$, the function $\chi \circ r$ on good classes of depth three is $\chi(\pm\pi_3(g))$ with a sign depending on g ($w^5 = \mathrm{N}(w)w$), which is not of the form $\chi' \circ \pi_3$ for any character χ', and $\Phi_1 \cdot (\chi \circ r)$ is not a virtual character of $\widetilde{U}_1$ for $\chi \neq \mathbf{1}$ (its restriction to $\widetilde{U}_3$ is λ_0 times $D \cdot \chi(\pm\pi_3)$, whose inner product with the linear character $\zeta^h \chi \circ \pi_3$ of H is not an integer). What is true is the statement for D_8-invariant characters, i.e. for restrictions of characters of $D^\times$, which is all that Target B needs; at level two the three-dimensional χ_3 happens to be invariant under the subgroup of D_8 that matters (χ_3 vanishes off the center of Heis, and $1+i$ acts trivially on that center), which is why (40) holds constituent-wise there. Second, for $m \ge 3$ the two statements (i), (ii) are both open. For (ii) the proof of Proposition 12.8 has an inductive flavor: $\hat{A}_m :=$ preimage of Q_{3m-2}/Q_{3m+1} is central in $\hat{H}_m$ for $m \ge 3$ without any hypothesis (for $x \in K^3$, $y \in K^{3m-2} \subset U_6$ one has $[\tilde{x}, \tilde{y}] = \ell([x,y]) \in \ell(K^{3m+1})$ because ℓ is a homomorphism on U_6 and Ad-equivariant), so a genuine irreducible μ of $\hat{H}_m$ has a central character $\hat{\nu}$ on $\hat{A}_m$, and $\mu \circ \mathrm{Ad}(t) = \mu$ forces $\alpha_\mu([t, A_m]) = 1$ for the character α_μ of $A_m = Q_{3m-2}/Q_{3m+1}$ with $\hat{\nu} = \lambda_0 \alpha_\mu$; freeness for the characters μ "of exact level m" (α_μ nontrivial on Q_{3m-1}/Q_{3m+1}) would follow from nondegeneracy properties of the brackets $Q_1/Q_2 \times Q_{3m-2}/Q_{3m-1} \to Q_{3m-1}/Q_{3m}$ and $Q_1/Q_2 \times Q_{3m-1}/Q_{3m} \to Q_{3m}/Q_{3m+1}$, and for the others from the lower level. We have not carried this out. For (i) at $m \ge 3$ the regular-character argument of Step 2 does not apply as such, because $\Theta_\rho \circ r$ no longer vanishes off a central subgroup, and a finite computation would have to replace Burnside's algorithm by Clifford theory relative to the abelian normal subgroup Q_4/Q_{3m+1} of H_m. Both are natural next steps.

13 Target C: uniform statements

This section records which parts of the argument are uniform in the odd prime p and the field F. Throughout: p is an odd prime, $F/\mathbb{Q}_p$ is finite with $\mu_p \subset F$ (so $(p-1) \mid e := e(F/\mathbb{Q}_p)$), $q = |k_F|$, ϖ is a uniformizer, $n = p$, E_0/F is the unramified quadratic extension, $D = E_0 \oplus E_0 j$ with $j^2 = \varpi$ and $jx = \sigma(x)j$, $K^m = 1 + \mathfrak{P}^m$, and

$$d := \frac{pe}{p-1}, \qquad k_0 := \frac{2e}{p-1} = \frac{2d}{p}, \tag{41}$$

so that $U_{E_0}^{(k)} \subset E_0^{\times p}$ for $k > d$ and $(U_{E_0}^{(i)}, U_{E_0}^{(j)})_p = 1$ for $i + j > d$ [FV, Ch. IV]; in the running example $p=3$, $e=2$, $d=3$, $k_0=2$. The cover is the restriction along ι of the Kazhdan–Patterson

cover of $\mathrm{GL}_2(E_0)$ with cocycle $\sigma_{E_0}(g,h)(\det g,\det h)_{E_0}^{c}$, σ_{E_0} as in (27) with the p-th power symbol, and $c \equiv -1/4 \pmod p$, which is the condition for scalars to be central (the computation of Lemma 10.1(b) gives $\beta(z,g)-\beta(g,z)=(1+4c)(z,\det g)$); for $p=3$ this is $c=2$. We write h_E for exponents of the p-th power symbol of E_0 and $(a,b)_E = h_E(a,b)$ additively.

Lemma 13.1 ([**P**] Depth facts)**.** *(a) For $k > k_0$ the p-th power map $K^k \to K^{k+2e}$ is bijective.*

(b) If $w \in D^\times$ and $w^p \in F^\times$, then $w \in F^\times$.

(c) Every element of K^{2d} is of the form zv^p with $z \in U_F^{(1)}$ and $v \in K^{k_0}$; z is unique modulo $F^{\times p}$ and $z \equiv \mathrm{Nrd}(X)^{(p+1)/2} \pmod{F^{\times p}}$.

Proof. (a) As in Lemma 7.1(a): $v_D(p)=2e$, and for $u \in \mathfrak{P}^k$ the terms $\binom{p}{i}u^i$ $(2 \le i \le p-1)$ and u^p lie in $\mathfrak{P}^{k+2e+1}$ when $k > k_0$. (b) If $w \notin F$ then $F(w)$ is a quadratic field in which $X^p - c$ ($c = w^p$) has a root; since $\mu_p \subset F$, $X^p - c$ is either split or irreducible over F, and $p > 2$ excludes the latter, so $w \in \mu_p w_0 \subset F$ for a root $w_0 \in F$. (c) Every $X \in K^{2d} \setminus F^\times$ lies in a unique maximal subfield E (and $X \in F^\times \cap K^{2d} = U_F^{(d)}$ is trivially of the required form). Recall the standard fact that for a local field $L \supset \mu_p$ with $e_L = e(L/\mathbb{Q}_p)$, the p-th power map $U_L^{(k)} \to U_L^{(k+e_L)}$ is bijective for $k > e_L/(p-1)$ [FV, Ch. I].

E ramified. Then $e_E = 2e$, $k_E = k_F$, $E \cap K^{2d} = U_E^{(2d)}$ (as $v_D = v_E$), and $U_F^{(d)} \subset U_E^{(2d)}$ induces an isomorphism $U_F^{(d)}/U_F^{(d+1)} \to U_E^{(2d)}/U_E^{(2d+1)}$ of residue groups ($\varpi_E^2 = \varpi\cdot$unit). Choose $z \in U_F^{(d)}$ with $z^{-1}X \in U_E^{(2d+1)}$. Since $k_0 + 1 > 2e/(p-1) = e_E/(p-1)$ and $k_0 + 1 + 2e = 2d+1$, the p-th power map $U_E^{(k_0+1)} \to U_E^{(2d+1)}$ is bijective, so $z^{-1}X = v^p$ with $v \in U_E^{(k_0+1)} \subset K^{k_0+1}$.

E unramified. Then $e_E = e$, $k_E = \mathbb{F}_{q^2}$, $E \cap K^{2d} = U_E^{(d)}$ (as $v_D = 2v_E$), and p-th powers of $U_E^{(k)}$ fill $U_E^{(k+e)}$ for $k > e/(p-1)$; the boundary level $k = e/(p-1) = k_0/2$ needs an argument. Write $\zeta_p = 1 + \varpi^{e/(p-1)}a_0 + \cdots$ with $a_0 \in \mathcal{O}_F^\times$ (possible since $v_F(\zeta_p - 1) = e/(p-1)$), so that $p/\varpi^e \equiv -\bar a_0^{\,p-1}$ in k_F (from $\zeta_p^p = 1$). For $u = 1+\varpi^{e/(p-1)}a$, $u^p \equiv 1 + \varpi^d(a^p + (p/\varpi^e)a)$ modulo $U_E^{(d+1)}$, so the induced map $k_E \to k_E$ on the graded pieces $U_E^{(e/(p-1))}/U_E^{(e/(p-1)+1)} \to U_E^{(d)}/U_E^{(d+1)}$ is $\bar a \mapsto \bar a^p - \bar a_0^{\,p-1}\bar a = \bar a_0^{\,p}\,\wp(\bar a/\bar a_0)$ with $\wp(x) = x^p - x$. Its image is $\bar a_0^{\,p}\ker(\mathrm{Tr}_{k_E/\mathbb{F}_p})$, of index p in k_E. The image of $U_F^{(d)}$ in $U_E^{(d)}/U_E^{(d+1)}$ is k_F, and $k_F \not\subset \bar a_0^{\,p}\ker(\mathrm{Tr}_{k_E/\mathbb{F}_p})$: for $u \in k_F$, $\mathrm{Tr}_{k_E/\mathbb{F}_p}(u/\bar a_0^{\,p}) = 2\,\mathrm{Tr}_{k_F/\mathbb{F}_p}(u/\bar a_0^{\,p})$, which is nonzero for suitable u because p is odd and the trace $k_F \to \mathbb{F}_p$ is onto. Hence k_F and the image together span k_E: choose $z \in U_F^{(d)}$ and $v_0 \in U_E^{(e/(p-1))}$ with $z^{-1}v_0^{-p}X \in U_E^{(d+1)} = (U_E^{(e/(p-1)+1)})^p$, say $= v_1^p$, and put $v = v_0 v_1 \in U_E^{(e/(p-1))} \subset K^{k_0}$ (as $v_D = 2v_E$).

In both cases $z \in U_F^{(d)} \subset U_F^{(1)}$ and $v \in E \cap K^{k_0}$. Uniqueness and the formula for z: $\mathrm{Nrd}\,X = z^2\,\mathrm{Nrd}(v)^p$, and squaring is bijective on $F^\times/F^{\times p}$, with inverse $y \mapsto y^{(p+1)/2}$. □

Theorem 13.2 ([**P**] The p-th power identity)**.** *Let $w = x + yj \in K^1$. Then $(w,0)^p = (w^p, 0)$ in the cover.*

Proof. If $y = 0$ the diagonal formula gives exponent $\sum_{k=1}^{p-1}\big(h_E(x^k,\sigma x) + 2ck\,h_E(\mathrm{N}\,x,\mathrm{N}\,x)\big) = \frac{p(p-1)}{2}h_E(x,\sigma x) = 0$. Let $y \neq 0$, $T = \mathrm{Trd}\,w$, $n = \mathrm{Nrd}\,w$, and define $S_k \in F$ by $S_0 = 0$, $S_1 = 1$, $S_{k+1} = TS_k - nS_{k-1}$, so that $w^k = S_k w - nS_{k-1}$ by Cayley–Hamilton, i.e. $w^k = x_k + y_k j$ with $y_k = S_k y$ and $x_k = S_k x - nS_{k-1}$. By Lemma 13.1(b), $S_k \neq 0$ for $1 \le k \le p$ ($S_k = 0$ would put $w^k \in E_0$ hence, as $w \notin E_0$, $w^k \in F$; for $k < p$ this forces $w \in F$ because $K^1F^\times/F^\times$ is pro-p, and for $k = p$ by (b)). In the cocycle $\beta(w^k, w)$ one has $x(\iota w^k) = \sigma(y)S_k$, $x(\iota w) = \sigma(y)$, $x(\iota w^{k+1}) = \sigma(y)S_{k+1}$, $\det \iota w^k = n^k$, so with $\tau = 1/S_k$ and $R = S_{k+1}/S_k$ (all in F), by the computation leading to (33),

$$\beta(w^k, w) = -(R,\tau)_E - (R, n^k)_E + c\,(n^k, n)_E = (S_{k+1}, S_k)_E - k(S_{k+1}, n)_E + k(S_k, n)_E,$$

using $(S_k,S_k)_E=(n,n)_E=0$. Put $a_k:=(S_{k+1},S_k)_E$ and $b_k:=(S_k,n)_E$ (so $a_1=b_1=0$). Summing over $k=1,\dots,p-1$ and collecting the b's,

$$\sum_{k=1}^{p-1}\beta(w^k,w)=\sum_{k=1}^{p-1}a_k+\sum_{k=2}^{p}b_k \pmod p. \tag{42}$$

The Cassini identity $S_{k+1}S_{k-1}-S_k^2=-n^{k-1}$ (induction on k) and the Steinberg relation for $u=S_{k+1}S_{k-1}/S_k^2$, $1-u=n^{k-1}/S_k^2$ (both nonzero for $2\le k\le p-1$) give, after expanding bimultiplicatively,

$$2(a_k-a_{k-1})=(k-1)\,(b_{k+1}-2b_k+b_{k-1})\qquad(2\le k\le p-1).$$

Hence $a_k=\frac12\sum_{j=2}^{k}(j-1)\Delta^2 b_j$, where $\Delta^2 b_j:=b_{j+1}-2b_j+b_{j-1}$, and interchanging the sums,

$$\sum_{k=1}^{p-1}a_k=\sum_{j=2}^{p-1}c_j\,\Delta^2 b_j,\qquad c_j:=\tfrac12(j-1)(p-j)\quad\text{(computations in }\mathbb{Z}[\tfrac12]/p).$$

Extend c_j to all integers j by the same quadratic formula; then $c_{j-1}-2c_j+c_{j+1}=-1$ for every j, $c_1=c_p=0$ and $c_0=c_{p+1}=-p/2\equiv 0$. Expanding $\Delta^2 b_j$ and reindexing,

$$\sum_{j=2}^{p-1}c_j\Delta^2 b_j=\sum_{j=1}^{p}b_j\,(c_{j-1}-2c_j+c_{j+1})-(c_0b_1+c_1b_2+c_pb_{p-1}+c_{p+1}b_p)\equiv-\sum_{j=1}^{p}b_j,$$

the four correction terms being those of the full double sum $\sum_{j=1}^{p}b_j(c_{j-1}-2c_j+c_{j+1})$ that do not occur in $\sum_{j=2}^{p-1}c_j\Delta^2 b_j$, each of which vanishes modulo p. Since $b_1=0$, (42) vanishes. For $p=3$ this is the computation of Proposition 10.2. □

Theorem 13.3 ([**P**] (H1) for every odd p)**.** *Define the star section on $U_{2d}=F^\times K^{2d}$ by $\ell(zv^p):=(z,0)(v,0)^p$ with $v\in K^{k_0}$. It is well defined, $\mathrm{Ad}(D^\times)$-equivariant, and a group homomorphism; explicitly $\ell(X)=(X,\kappa(X))$ with*

$$\kappa(X)=-\tfrac{p+1}{2}\,(y_X,\operatorname{Nrd}X)_E\quad(y_X\neq0),\qquad\kappa(X)=0\quad(y_X=0).$$

Consequently, for θ_0 with $\theta_0^p=1$ and $\widetilde\theta(\zeta,z)=\zeta\theta_0(z)$, the analogue ψ_{2d} of ψ_6 ($\zeta\widetilde\theta(\tilde z)$ at $\zeta\tilde z v^*$) is a character of $\widetilde U_{2d}$, given on $\widetilde K^{2d}$ by

$$(X,e)\longmapsto\zeta^{\,e-\kappa(X)}\,\theta_0(\operatorname{Nrd}X)^{(p+1)/2},$$

since the value at the star lift $\ell(X)$ is $\theta_0(z)$ with $z\equiv\operatorname{Nrd}(X)^{(p+1)/2}$ (Lemma 13.1(c)). For $p=3$ the exponent $2\equiv-1$ recovers (31).

Proof. Theorem 13.2 gives $\ell(zv^p)=(zv^p,\beta(z,v^p))$, and $\beta(z,g)=(z,x(g))_E+2c(z,\det g)_E$ with $\det g=\operatorname{Nrd}(v)^p$, so $\kappa(X)=(z,y_X/z)_E=(z,y_X)_E=\frac{p+1}{2}(\operatorname{Nrd}X,y_X)_E$ by Lemma 13.1(c) (and 0 if $y_X=0$, as $(z,x_X)_E=\frac{p+1}{2}(x_X\sigma x_X,x_X)_E=0$ for $x_X\in U^{(d)}$). Well-definedness is the uniqueness in Lemma 13.1(c) together with $h_E(F^{\times p},\cdot)=0$. For multiplicativity we repeat the proof of Proposition 10.4 with $m:=-\frac{p+1}{2}$, so $2m\equiv-1$, for $X=a+bj$, $X'=a'+b'j\in K^{2d}$: now $a,a'\in U^{(d)}$, $b,b'\in\varpi^d\mathcal O$, $n,n'\in U_F^{(d)}$, all mutual symbols among $a,\sigma a,a',\sigma a',n,n'$ vanish ($2d>d$), the twist $c(n,n')_E$ vanishes, $n\equiv\mathrm{N}\,a$ modulo $U^{(2d+1)}\subset E_0^{\times p}$, and (28)–(32) hold with $U^{(3)}$ replaced by $U^{(d)}$ (as $1+d>d$ and $\mu_{q^2-1}\subset E_0^{\times p}$), with $\lambda(a)=(\varpi,\mathrm{N}\,a)_E=2(\varpi,a)_E$. In the main case (30) becomes

$(R,\tau)_E = m(\tau,n')_E - (m+1)(R,n)_E - m(R,n')_E = -v(\tau)(\varpi,a')_E - v(R)(\varpi,a)_E + v(R)(\varpi,a')_E$, which is exactly the value of $(R,\tau)_E$ computed from $R = a'(1+\tau_2)$; in the cases $b=0$, $b'=0$, $b=b'=0$ and $R=0$ the required identities are $(2m+1)v(b')(\varpi,a)_E = 0$, $2mv(b)(\varpi,a')_E = -v(b)(\varpi,a')_E$, $0=0$ and $(2m+1)v(b)[(\varpi,a)_E+(\varpi,a')_E]=0$ respectively, all true since $2m+1\equiv 0$. The passage from K^{2d} to U_{2d} and the formula for ψ_{2d} are as in Corollary 10.3(d). □

Remark 13.4 (What is and is not uniform)**.** (i) **[P]** Uniform in p and F (for $n=p$): the toral theorem (Section 4); the positivity criterion; the cocycle model and the p-th power identity; the depth facts; hypothesis (H1) in the form of Theorem 13.3; the product formula and the criterion of Section 12 (with 3 replaced by p, K^3 by K^p and λ_0 by the character produced by the analogue of Theorem 9.6, once available).

(ii) **[O]** Expected to be uniform but not written: the Brauer reduction. $D^\times/F^\times K^1$ is now the group of order $2(q+1)$ generated by $k_{E_0}^\times/k_F^\times$ and j; the centralizer lemma uses only that p is odd; the ℓ-elementary subgroups for $\ell \neq p$ live in tori or in the preimage of a 2-subgroup, and the 2-group lemma needs the same analysis as in Theorem 12.1, with SD_{16} replaced by the appropriate extension.

(iii) **[C]** Not uniform: the finite design. The good elements of depth exactly p (the analogue of G_3, with leading coefficients $w^{1+(p-1)(q+1)/2}$, which need not exhaust $k_{E_0}^\times$ when $\gcd(1+(p-1)(q+1)/2,\, q^2-1)>1$) and the quotient $V = Q_p/Q_{2d}$ depend on (p,e,q), and the uniqueness of the defect function was a computation. A conceptual replacement for Section 9 is the main open problem of Target C; the next test cases are $F=\mathbb{Q}_3(\zeta_3,\sqrt{3})$ $(e=4,\, d=6)$ and $F=\mathbb{Q}_5(\zeta_5)$ $(e=4,\, d=5)$.

(iv) For composite odd n and for $p\nmid n$ nothing new is needed at tame places (Part III); for $n=p^a m$ with $a\geq 2$ the depth bookkeeping changes (n-th powers of K^k land in K^{k+2ae}) and we have not examined it.

13.1 Proof checkpoints retained from Draft 0

A positive solution for a row must still supply (i) an explicit cocycle/central-character normalization (now available, uniformly in p), (ii) a finite-level factorization or an induced construction (now: Ind from $\widetilde{U}_3$, Proposition 12.4), (iii) exact full-group character coefficients (now: the Fourier identities of Proposition 9.4), and (iv) a uniform all-ρ argument (open beyond level one).

14 Interface with the larger Jacquet–Langlands program

The companion problem is local. Theorem 11.1 and Corollary 11.4 remove, for the rows they cover, the "L^2-unitarity-to-integrality" gap recorded in Part III, Draft 9, at the wild place $\mathfrak{p}_3$ of the running example; in particular the character relation (D2) of Part I now has, at $\mathfrak{p}_3$, a proved instance for $\rho=\mathbf{1}$ and the level-zero rows. Even full success on Target B would not on its own prove the additional global comparison of ordinary $D^\times$ and the covered group: that comparison remains conditional on matching functions at exceptional split places, complex-place normalization, a global orbital-coefficient identity, and the support/germ hypotheses stated in Part IV. Conversely, the global comparison could be established in a restricted test range without first settling every local irreducible-character row. These projects should be developed in parallel but not conflated.

A Tables

Coordinates. $V = Q_3/Q_6 \simeq \mathbb{F}_3^5$ with basis $(A_{\mathrm{re}}, A_{\mathrm{im}}, C_{\mathrm{im}}, B_{\mathrm{re}}, B_{\mathrm{im}})$ as in (19); a vector is written as a string of five digits. The generators act by the matrices

$$\mathrm{Ad}(1+j):\begin{pmatrix}1&0&0&0&0\\0&1&0&0&0\\0&1&1&0&0\\0&0&0&1&0\\0&2&1&0&1\end{pmatrix},\quad \mathrm{Ad}(1+ij):\begin{pmatrix}1&0&0&0&0\\0&1&0&0&0\\2&0&1&0&0\\2&0&2&1&0\\0&0&0&0&1\end{pmatrix},\quad \mathrm{Ad}(1+i):\begin{pmatrix}0&2&0&0&0\\1&0&0&0&0\\0&0&1&0&0\\0&0&0&0&2\\0&0&0&1&0\end{pmatrix},\quad \mathrm{Ad}(j):\begin{pmatrix}1&0&0&0&0\\0&2&0&0&0\\0&0&2&0&0\\0&0&0&1&0\\0&0&0&0&2\end{pmatrix}.$$

The 36 good lines and the defect function e_*. Each entry is a representative v followed by $e_*(v)$; $e_*(-v) = -e_*(v)$.

01002:0	01012:0	01022:0	01101:0	01111:1	01121:2
01201:0	01211:2	01221:1	10020:0	10021:0	10022:0
10110:0	10111:2	10112:1	10210:0	10211:1	10212:2
11002:0	11011:0	11020:0	11101:1	11110:2	11122:0
11201:2	11210:1	11222:0	12001:0	12012:0	12020:0
12102:2	12110:1	12121:0	12202:1	12210:2	12221:0

The 27 covectors $\Lambda = \Lambda_+$ (those h with $N_{e_*}(h) = 20$):

00000	00100	00200	01001	01121	01211	02002	02112	02222
10010	10111	10212	11022	11110	11201	12021	12102	12210
20020	20122	20221	21012	21101	21220	22011	22120	22202

The Heis-cocycle attached to e_* is $\beta_{\mathrm{Ad}(1+j)} = (1,0,0,1,0)$, $\beta_{\mathrm{Ad}(1+ij)} = (0,1,0,0,1)$. The numbers of ordered coplanar triples of distinct good lines is $B = 720$, all of them defect free.

The Hilbert pairings. On the basis $\varpi,\ 1+\varpi,\ 1+\varpi i,\ 1+\varpi^2,\ 1+\varpi^2 i,\ 1+\varpi^3$ of $E_0^\times/E_0^{\times 3}$ (the last five being a basis of $U_{E_0}^{(1)}/U_{E_0}^{(1)3}$; recall $1+\varpi^3 i$ is a cube) the cubic Hilbert pairing of E_0, written as exponents of ζ, is antisymmetric with upper-triangular entries

$$(\varpi, 1+\varpi^3) = 2,\quad (1+\varpi, 1+\varpi i) = 1,\quad (1+\varpi, 1+\varpi^2) = 2,\quad (1+\varpi i, 1+\varpi^2 i) = 1,$$

and all other upper-triangular entries 0. On the basis $\varpi, 1+\varpi, 1+\varpi^2, 1+\varpi^3$ of $F^\times/F^{\times 3}$ the pairing of F has $(\varpi, 1+\varpi^3) = 1$, $(1+\varpi, 1+\varpi^2) = 1$ and all other upper-triangular entries 0. Both are nondegenerate, Galois invariant, and satisfy $(a,b)_{E_0} = (a, \mathrm{N}\, b)_F$ for $a \in F$; the common sign is the choice of the identification $\mu_3 \simeq \mathbb{Z}/3$.

B Verification scripts

All scripts are in pure Python with exact rational or modular arithmetic, with two exceptions: script 11 uses floating point only to display complex character values, and scripts 13–14 obtain the character table of Q_3/Q_7 by Burnside's algorithm in floating point and then round the values to $\mathbb{Z}[\zeta_3]$ (maximal rounding error $2 \cdot 10^{-13}$) and re-verify all orthogonality relations exactly; this is numerical discovery followed by exact verification, and since Draft 6 those two scripts are corroboration only (Theorem 12.7 is proved by hand). They are kept with the paper under `part-IV/scripts/`.

1. `step1_G3.py`: exact arithmetic in $\mathcal{O}_D$; the 648 depth-one classes modulo $K^4 U_F^{(1)}$; their cubes in V; the 72 elements of $\bar{G}_3$, the 36 lines, agreement with (20).

2. `step2_finite.py`: the finite group Q/Q_7 (3^9 elements); the cube map; depths of cubes (3 or ≥ 6); $\mathrm{Cubes}(Q_2) = Q_6$; Q_6-invariance of G_3.

3. `step3_actions.py`: the actions of $\mathrm{Ad}(1+j), \mathrm{Ad}(1+ij), \mathrm{Ad}(1+i), \mathrm{Ad}(j)$ on V; orders of the groups they generate (27, 8, 216); orbits on $\bar{G}_3$ and on the lines; $B = 720$.

4. `step4_design.py`, `step6_orbit.py`: the design search (the nine, resp. 3^{10}, candidates), the cocycle extension test, the simply transitive orbit, the Fourier identities.

5. `step5_H1.py`: (a) the Lazard form on K^4/K^8: invariant alternating forms vanishing on torus subspaces are 0; (b) odd functions on Q_6/Q_8 with equivariant defects are linear.

6. `cover.py`, `step7_symbols.py`: Steinberg relations and the Hilbert symbols of F and E_0; nondegeneracy, Galois invariance, norm compatibility; the cocycle identity for (27) (250 triples); the diagonal formula; triviality on scalars; the torus commutator formula of Part I (200 pairs).

7. `step8_H1test.py`, `step9_H1_stratified.py`, `step12_H1_proof_check.py`: corroboration of Proposition 10.4: (H1) through cube roots (Hensel in $F(X)$), through (30) (7600 stratified instances), and through (34) and (30) on 12,000 instances covering every case of the proof; script 12 also verifies the closed form (29) against the digit form and the symbol facts (28).

8. `step10_defects.py`: defects of 3- and 4-term relations in $\bar{G}_3$ against e_*; $\omega = 0$; $\kappa_{1,5}$.

9. `step11_endtoend.py`, `step11b.py`: the character of $\mathrm{Ind}_{\widetilde{U}_3}^{\widetilde{U}_1} \lambda_0$ from (25) against (21).

10. `step13_level2.py`, `zsolve.py`: the group $H = Q_3/Q_7$ (multiplication table, 297 classes), its character table by Burnside's algorithm with exact re-verification, the $\mathrm{Ad}(Q_1)$-orbits on $\mathrm{Irr}(H)$, the 219 good classes and the cube root; the criterion of Proposition 12.4 tested by exact integer linear algebra (Smith normal form) for the four D_8-orbit sums of Irr(Heis).

11. `step14_level2_restriction.py`: the decomposition of $(\chi \circ r)D$ into irreducible characters of H for the orbit sums and for χ_3, χ_3'.

12. `step15_level2_clifford.py`: λ_0 on $\hat{H}$ in the cocycle model, the twisted action of Q_1/Q_3 on the genuine irreducible characters of $\hat{H}$, an explicit stabilizer check (no non-trivial coset representative fixes any genuine character), and the orbit table written to `level2_orbit_certificate.txt` (two orbits of size 27 on the degree-three characters; nine of size 27 on the linear ones). Running time about ten minutes. Corroborates Proposition 12.8.

Acknowledgments and research-status note

The computations stem from the collaborative Part IV project of D. Joyner, ChatGPT (OpenAI), and Claude (Anthropic). Claude's referee report on Part IV, Draft 5 suggested the positivity criterion and a uniform rank-two toral analysis; the reduction to the Sylow 3-subgroup, the finite model, the explicit cocycle model and the verification scripts of Drafts 1–3 were produced by Claude; the proof of Proposition 10.4 in Draft 2 replaces a case-by-case argument of Draft 1 whose first case was incorrect, as pointed out in ChatGPT's referee report on Draft 1. Sources labeled [E] are imported from Parts I–III or [JC]; results labeled [P] are proved here; results labeled [P^c] rely in addition on the exhaustive exact computations listed in Appendix B and itemized in Remark 11.2. Independent refereeing of the hand arguments and re-execution of the scripts are advisable before the result is cited.

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